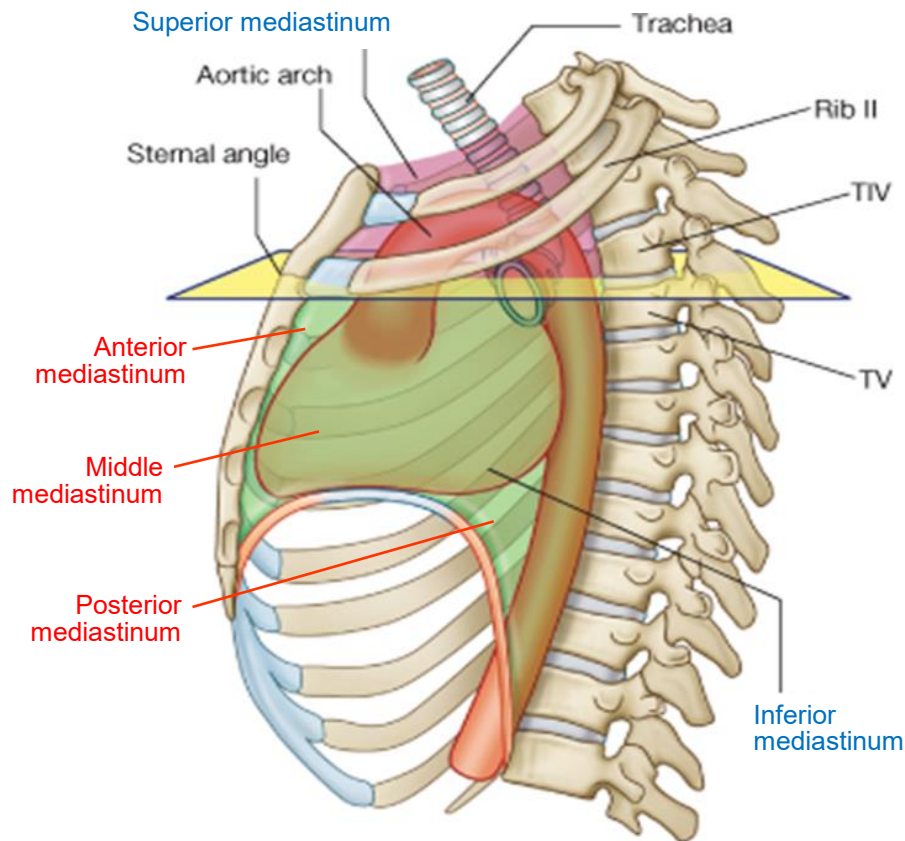




Lab Gross Anatomy of the Heart and Great Vessels

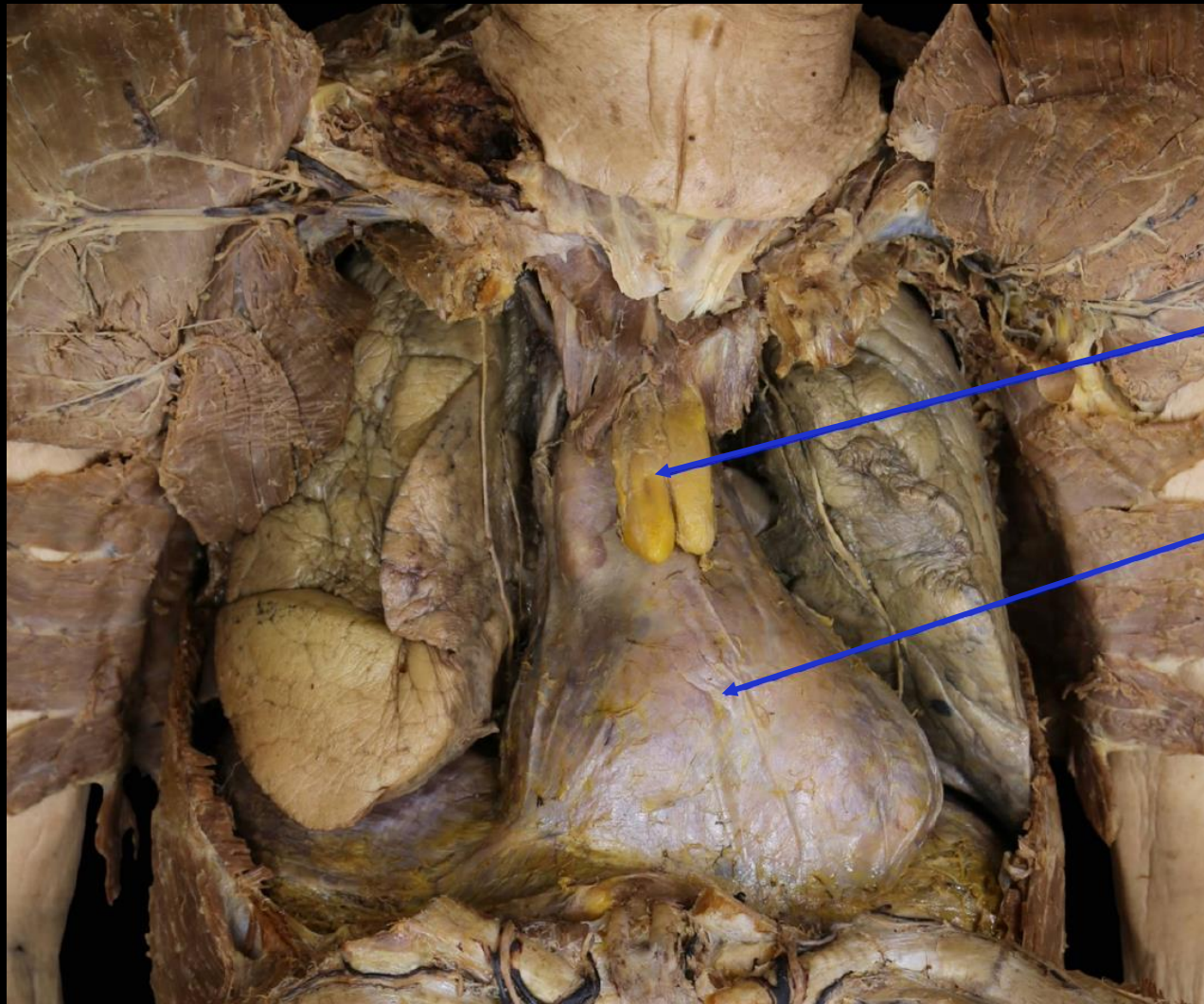
Assoc.Prof.Dr. Pornsawan Duangsuwan

Mediastinum



- Superior mediastinum
- Inferior mediastinum
 - Anterior mediastinum
 - Middle mediastinum
 - Posterior mediastinum

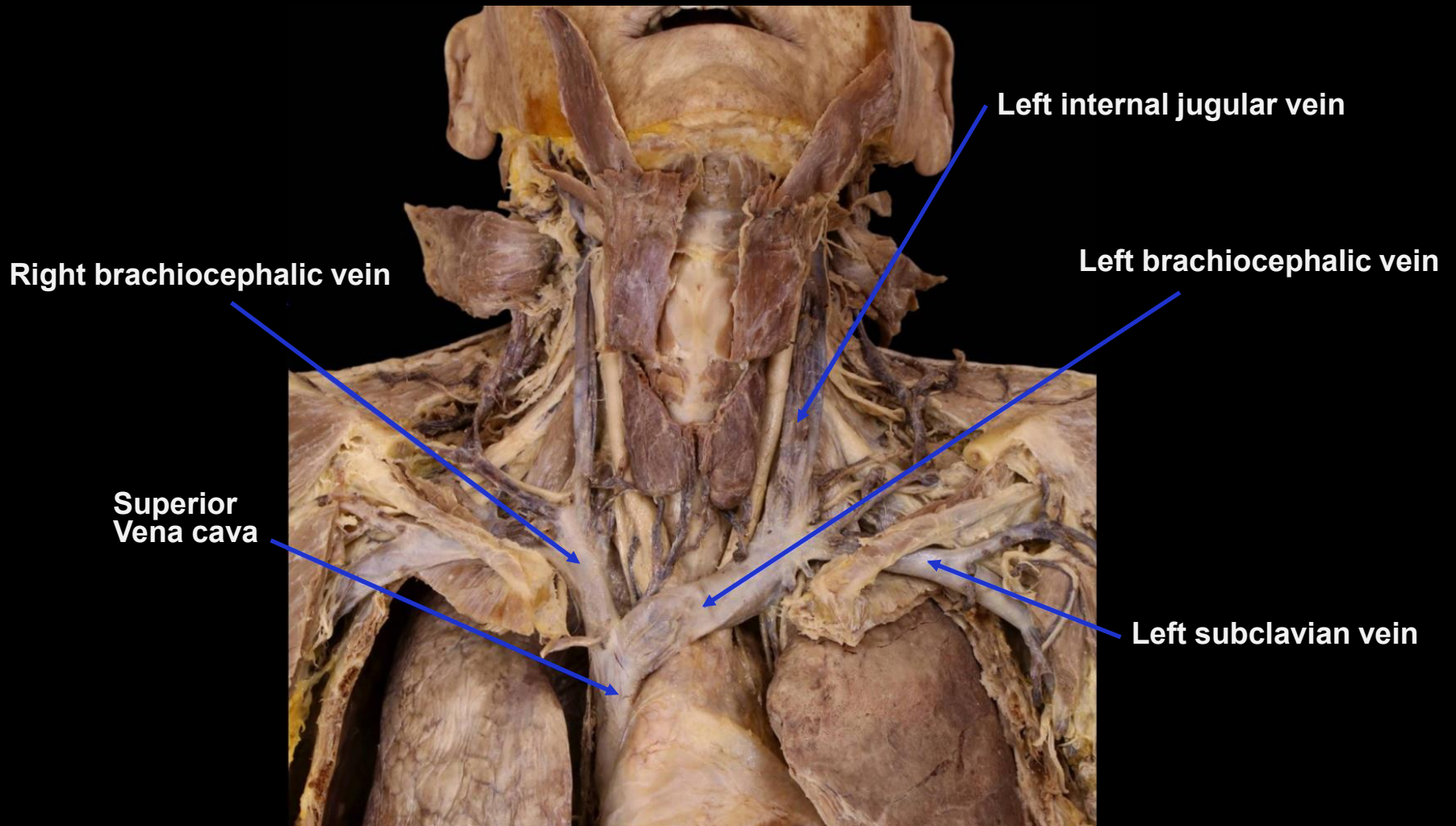
Superior & Anterior mediastinum



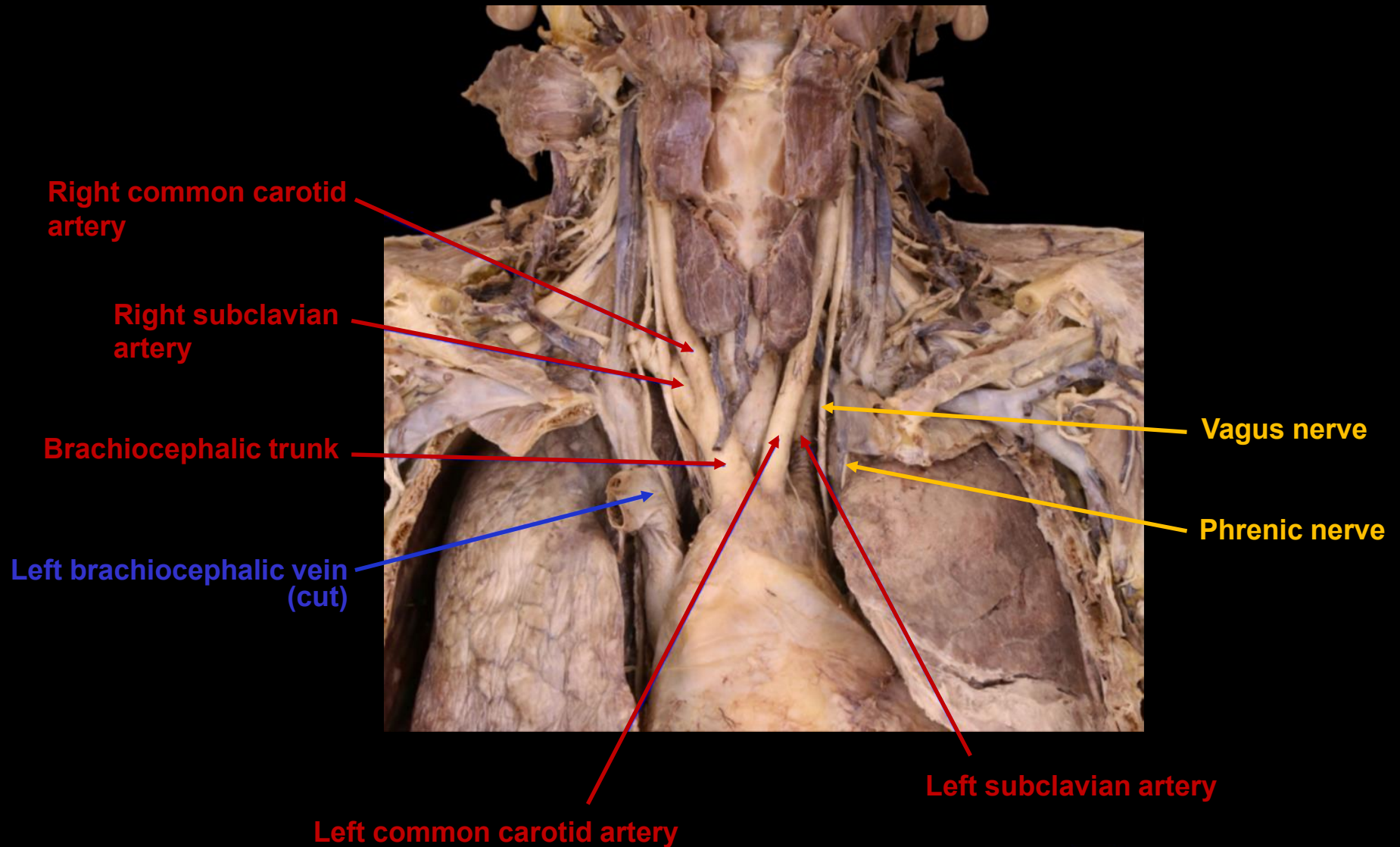
**Thymus
gland**

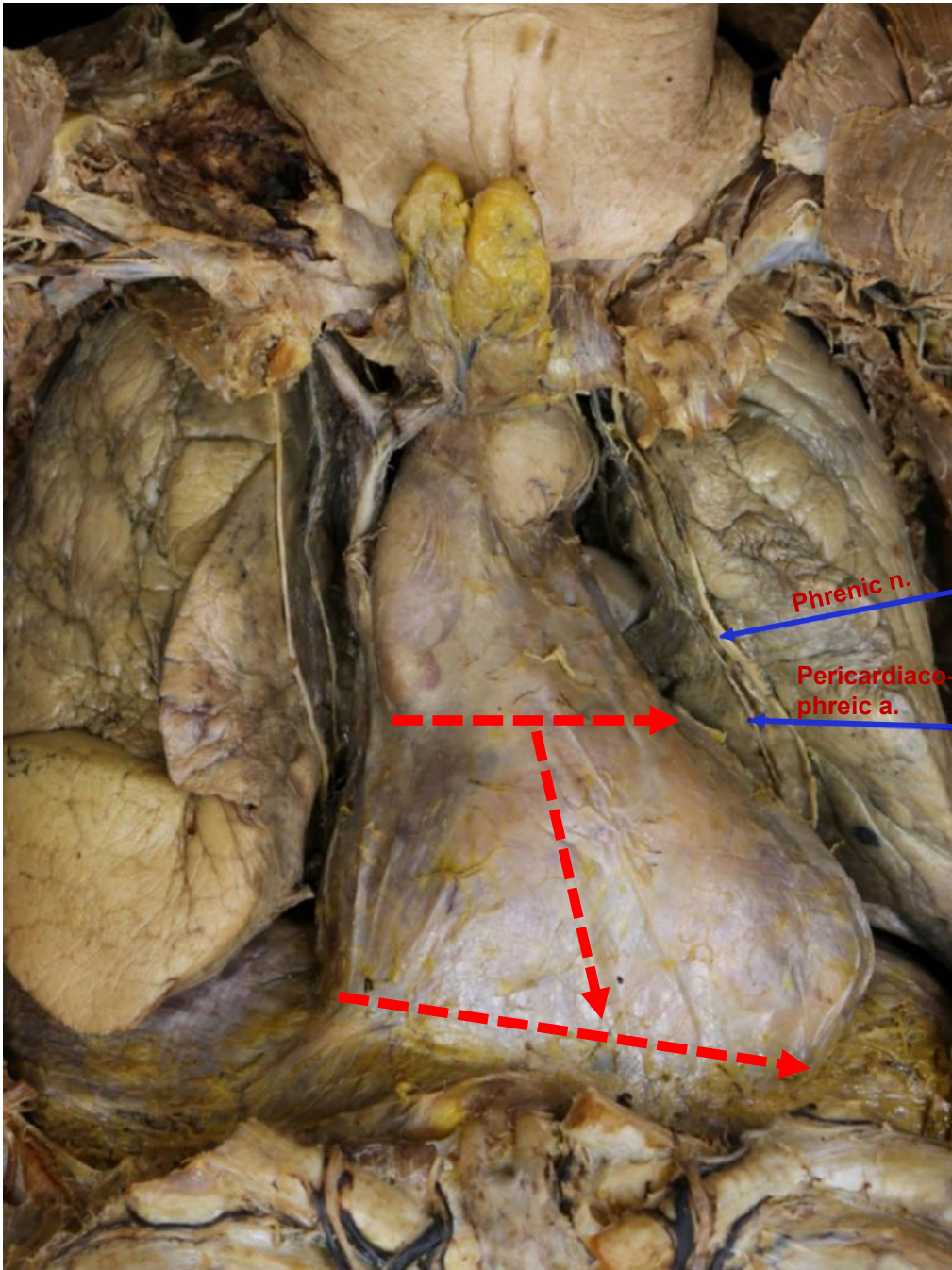
**Fibrous
pericardium**

Superior mediastinum



Superior mediastinum

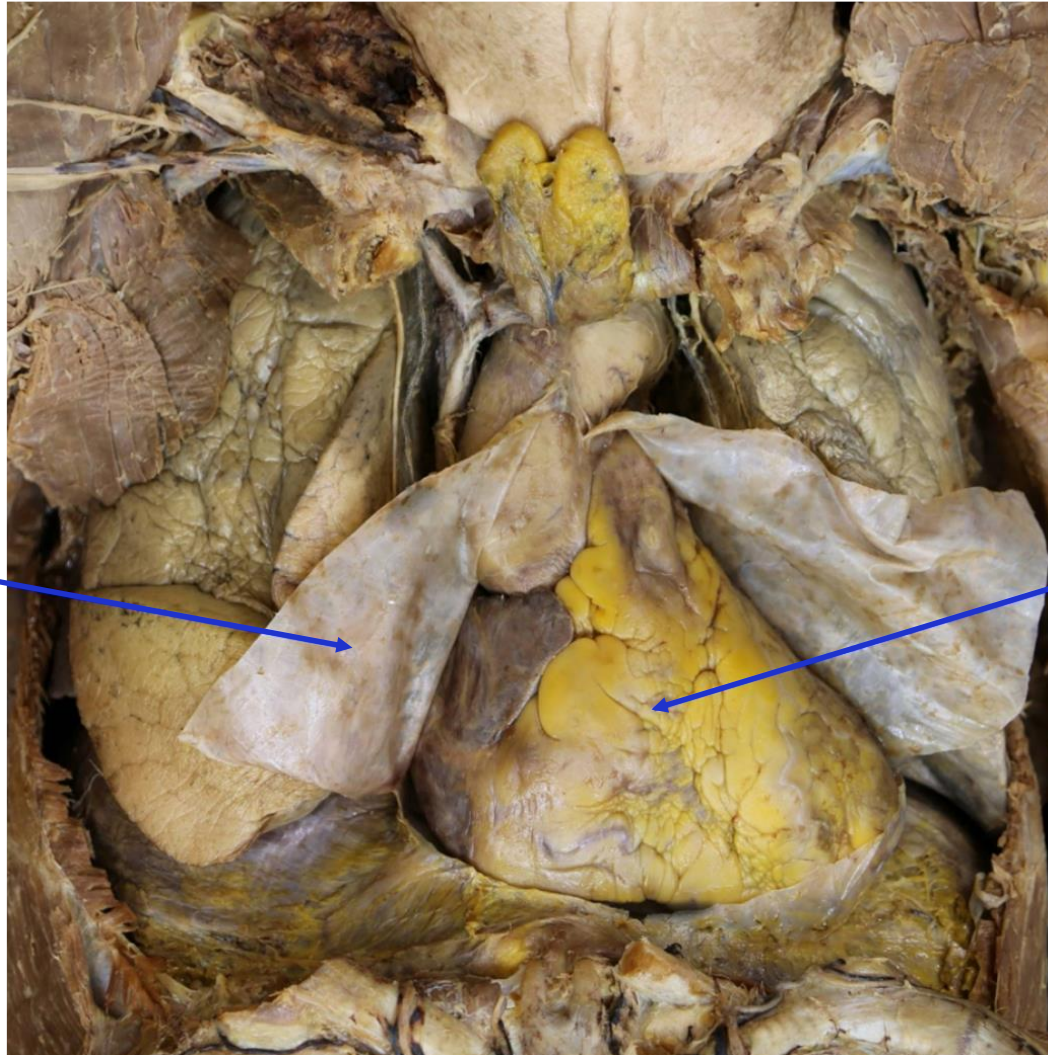




Open the pericardial sac by making an I incision

1. Horizontal incision close to the attachment of the pericardium to the diaphragm
2. Horizontal incision approximately at the level where the great vessels of the heart begin
3. An oblique incision connecting the two horizontal incisions
4. Grab the pericardial flap with forceps and remove it.

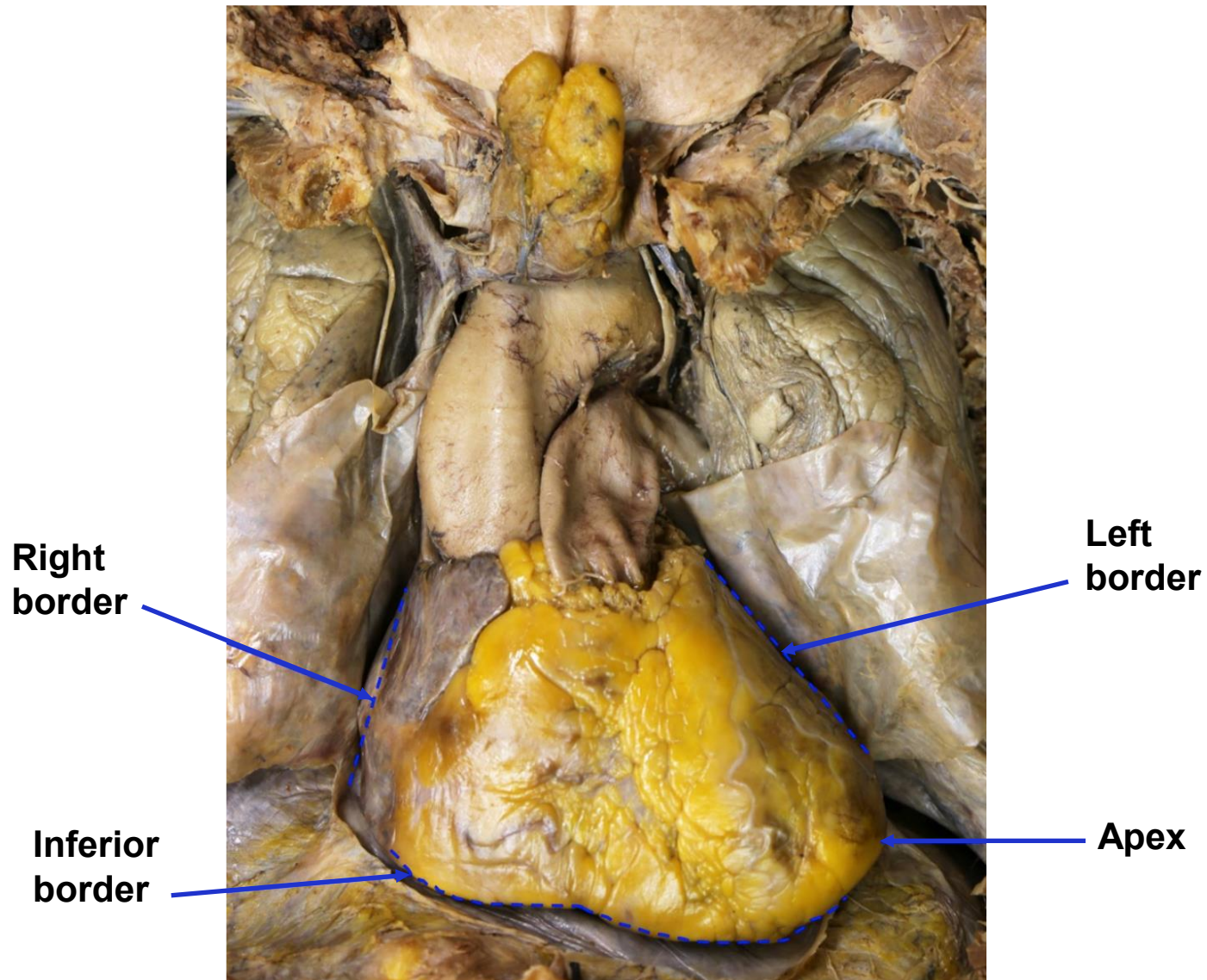
The pericardium has been reflected to expose the heart



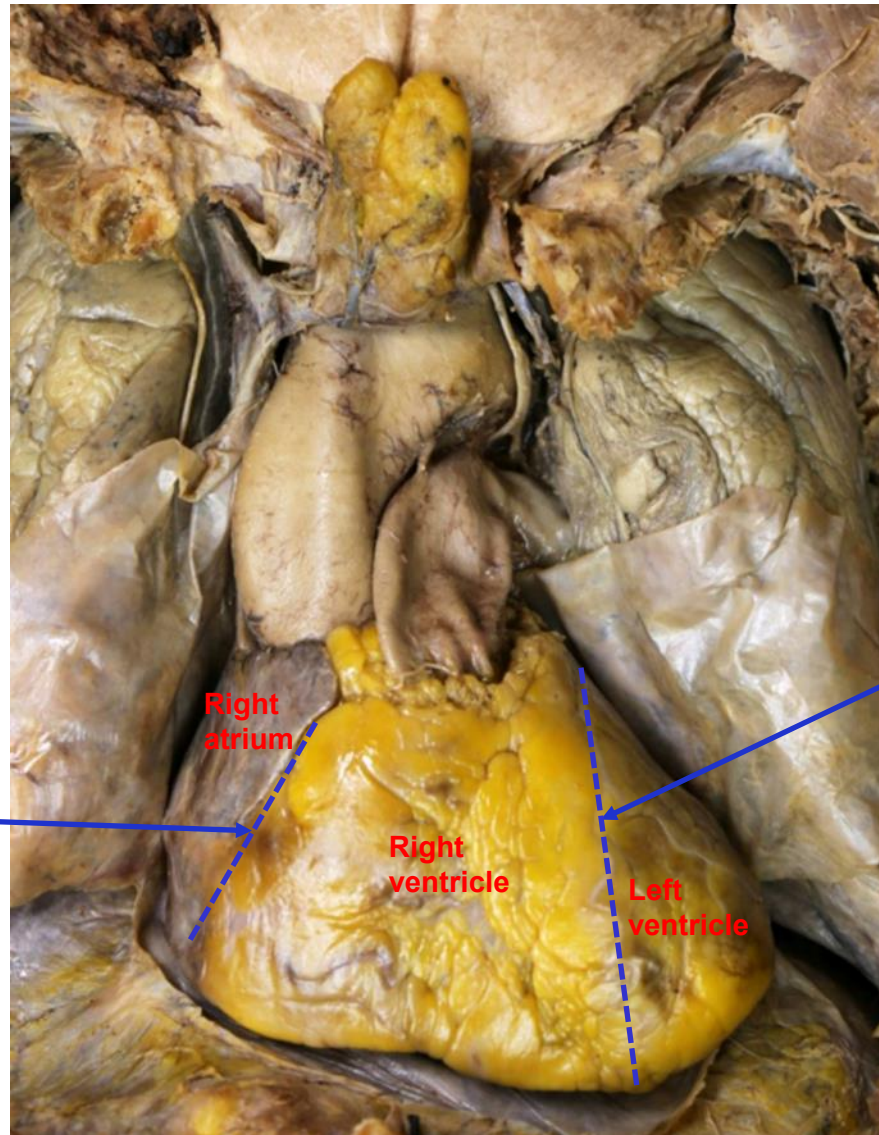
**Parietal layer of
serous pericardium**

**Visceral layer of
serous pericardium**

Examine the gross features of the heart and great vessels



Examine the gross features of the heart and great vessels



**Atrioventricular sulcus
(coronary sulcus)**

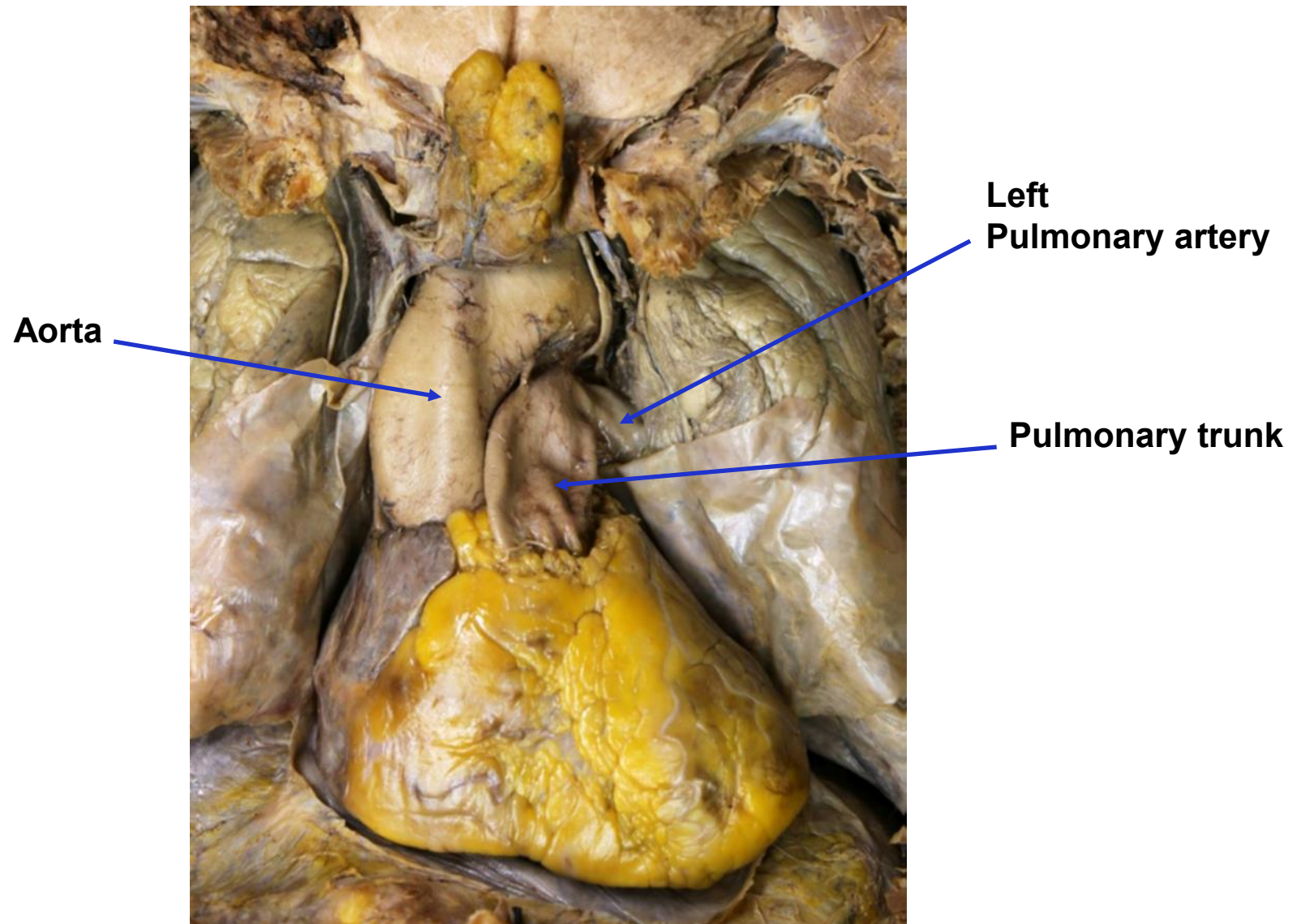
**Right
atrium**

**Right
ventricle**

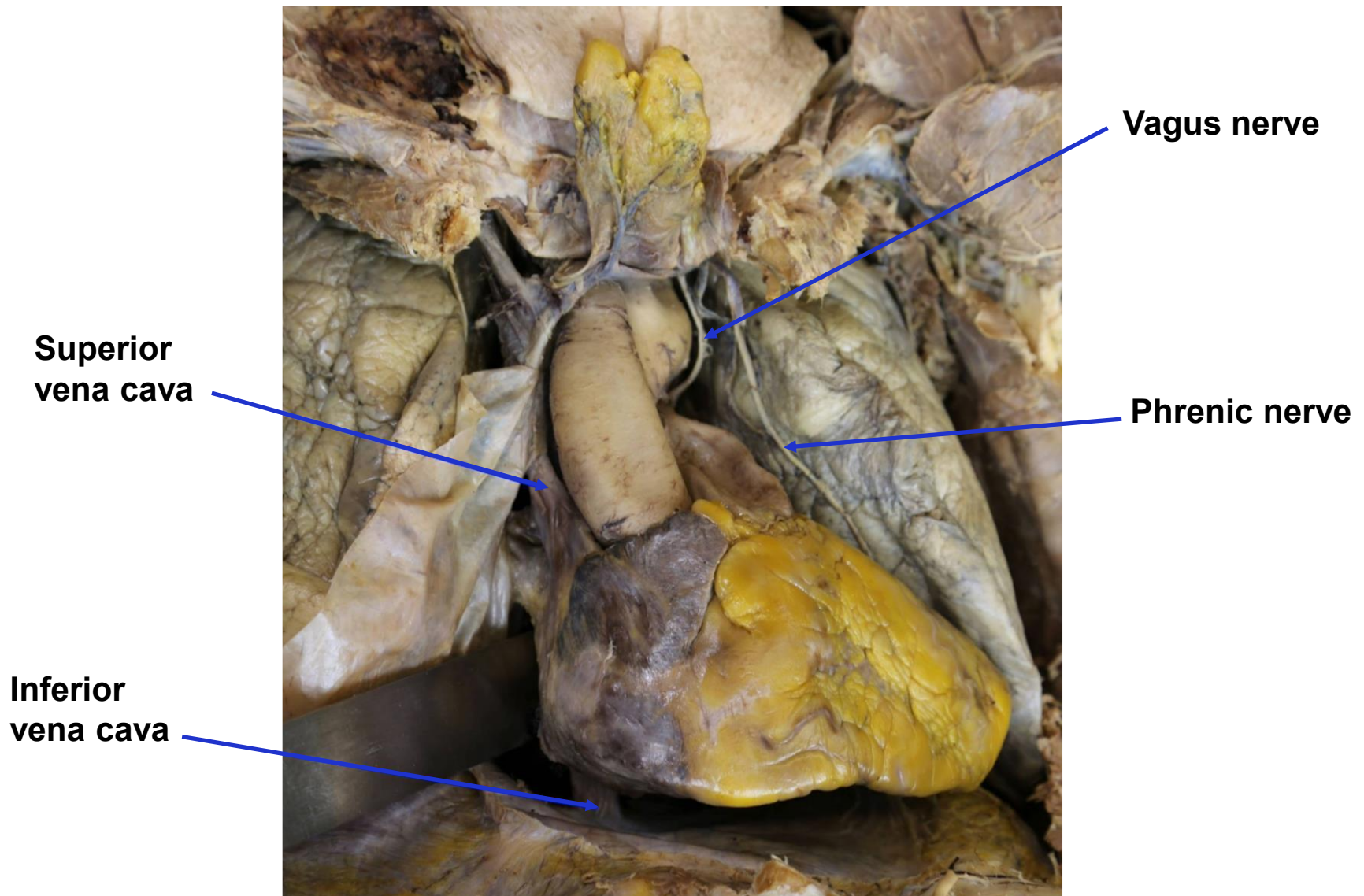
**Left
ventricle**

**Anterior
Interventricular sulcus**

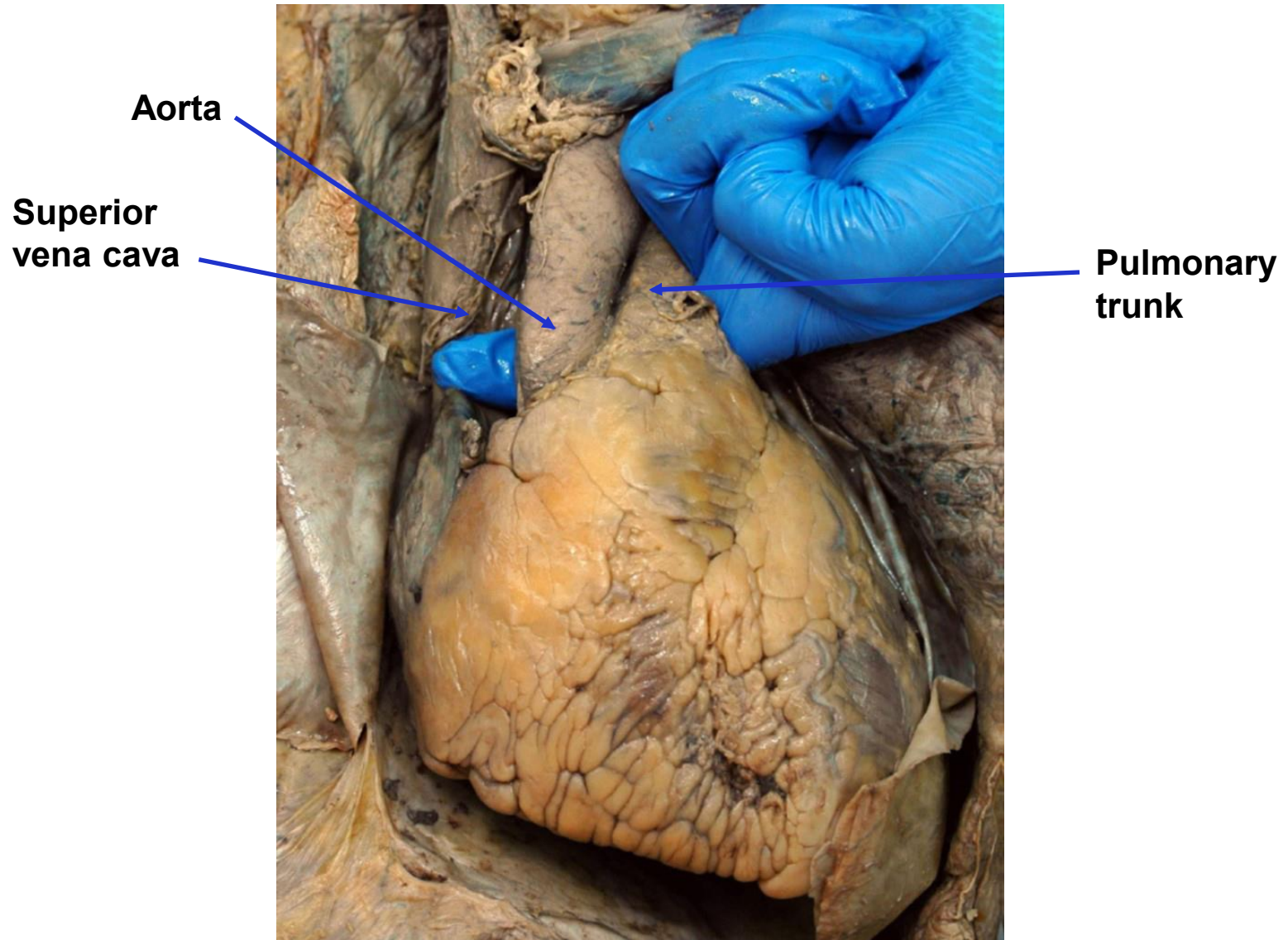
Examine the gross features of the heart and great vessels



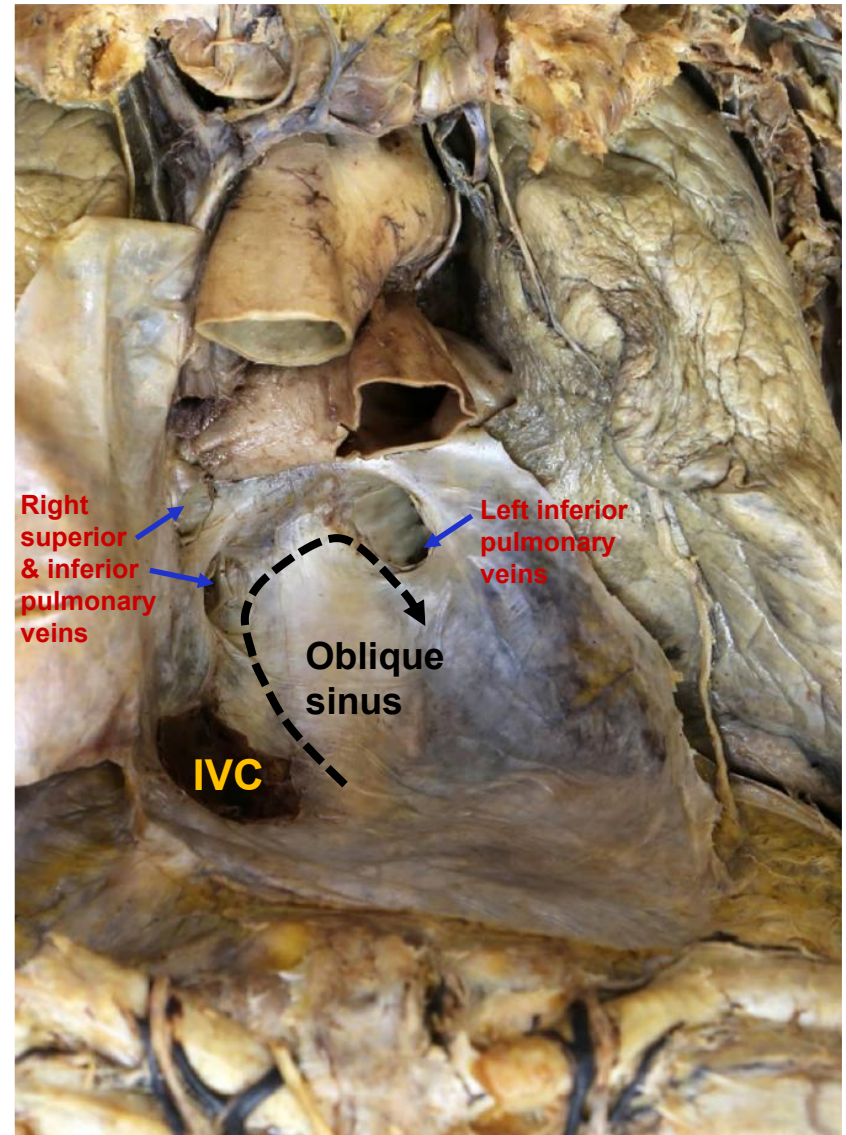
Examine the gross features of the heart and great vessels



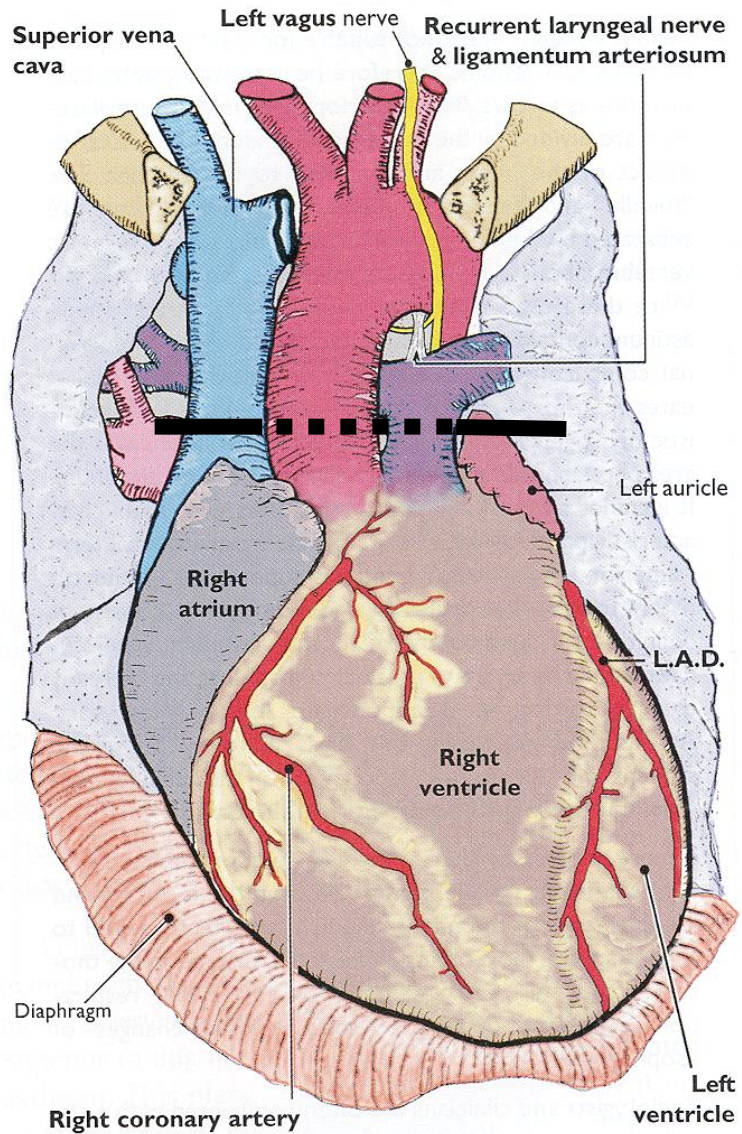
Examine the transverse sinus



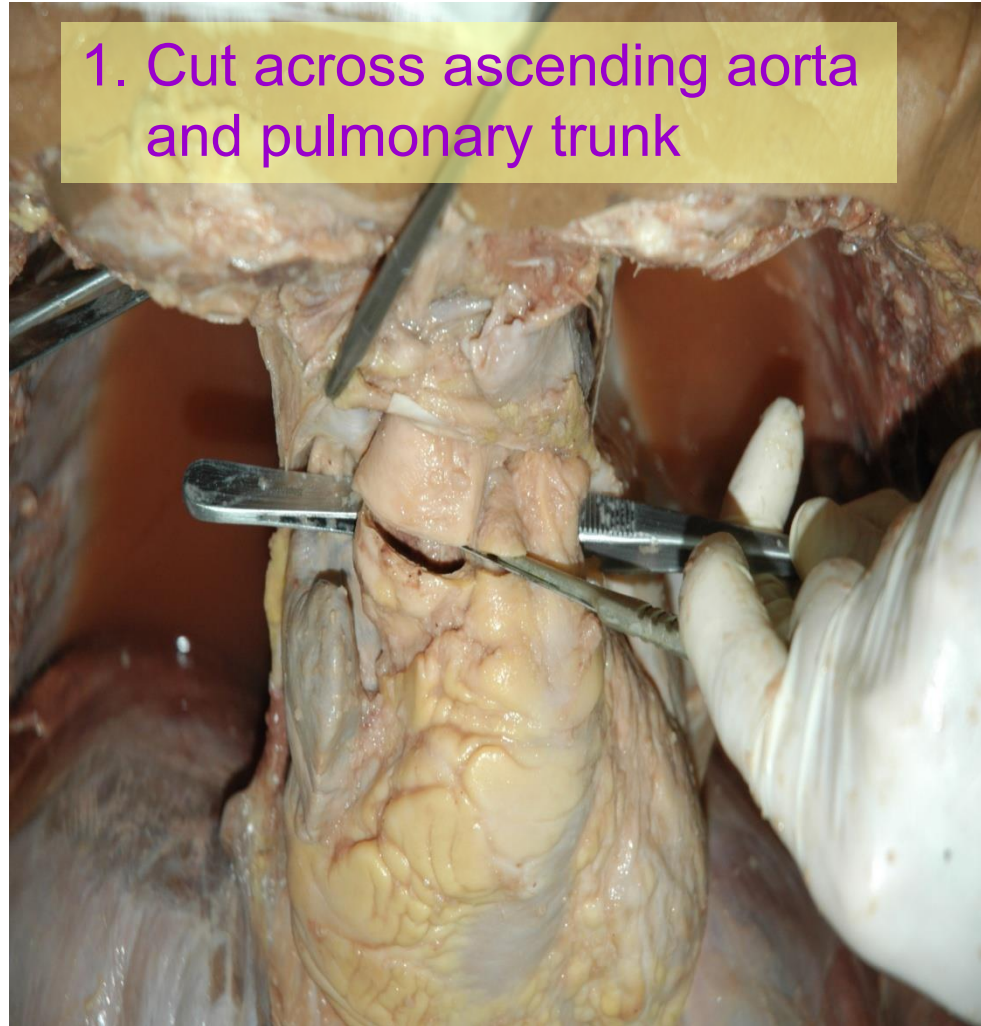
Examine the oblique sinus



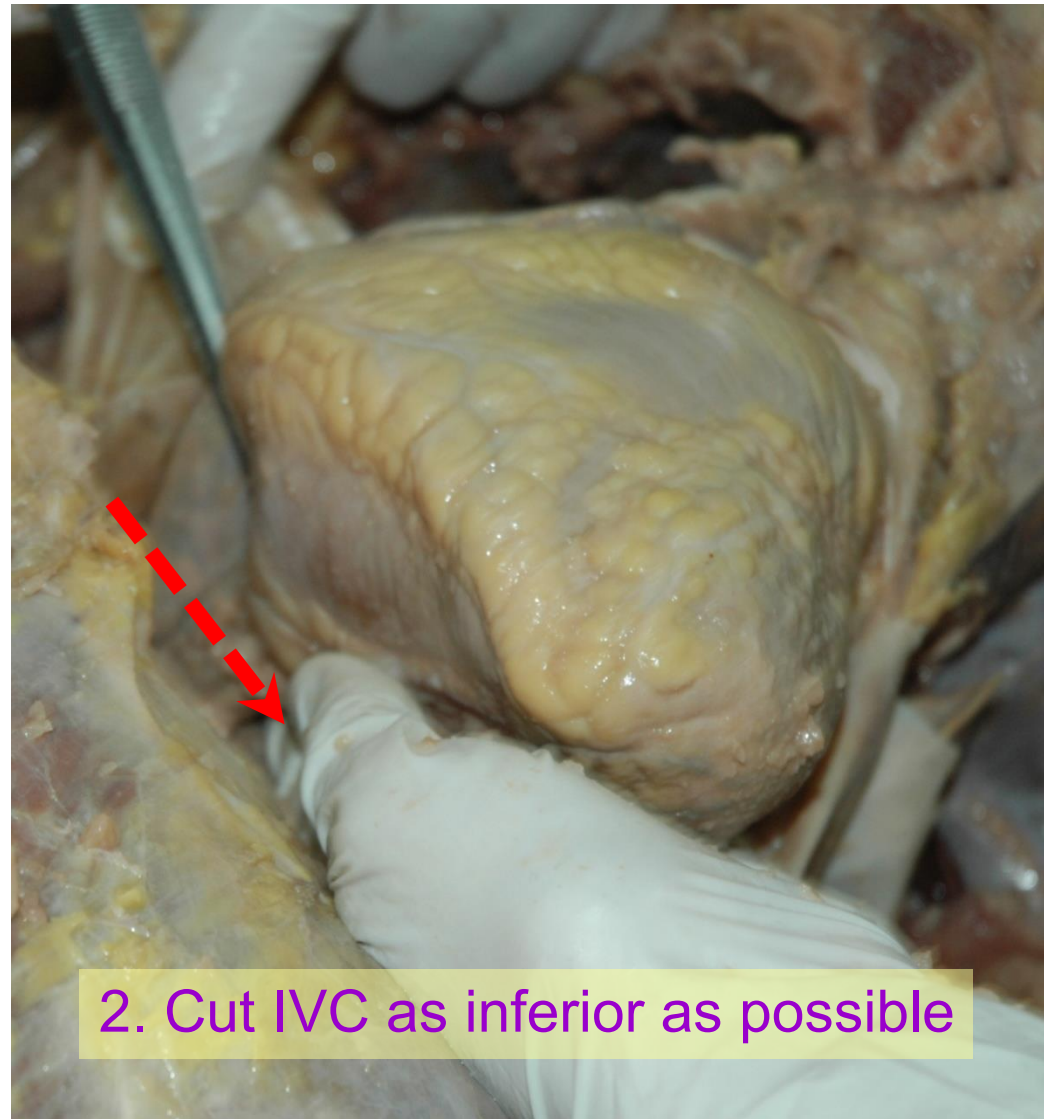
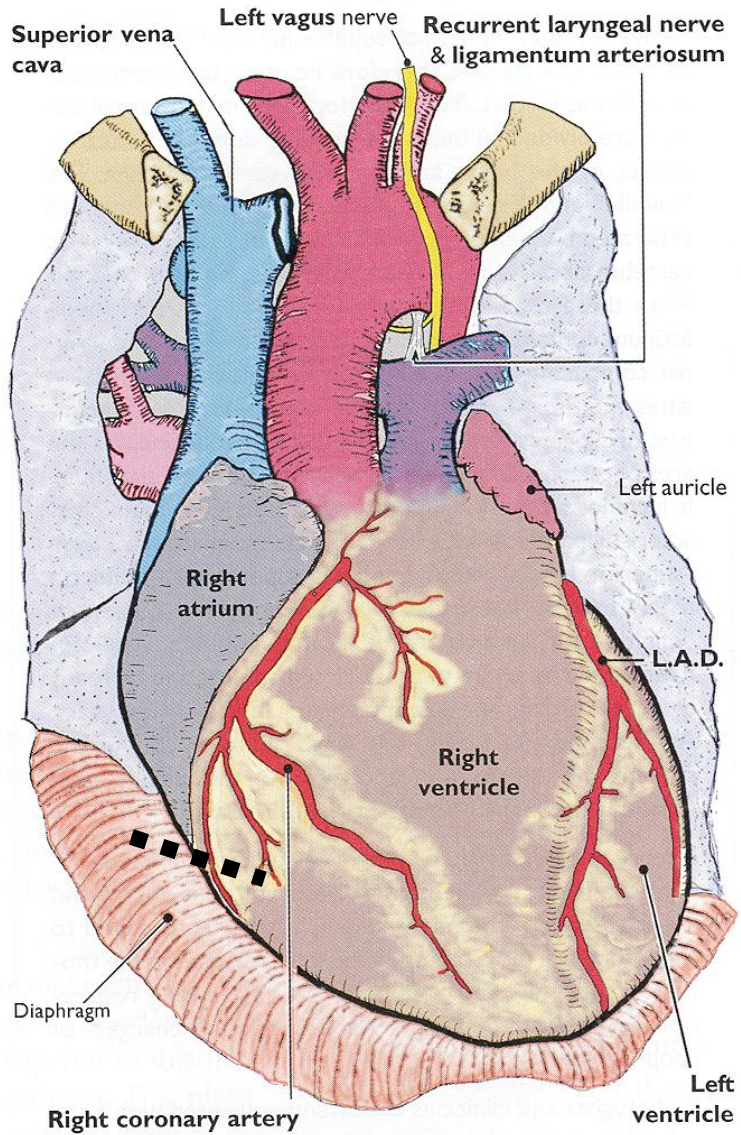
Removal of the heart



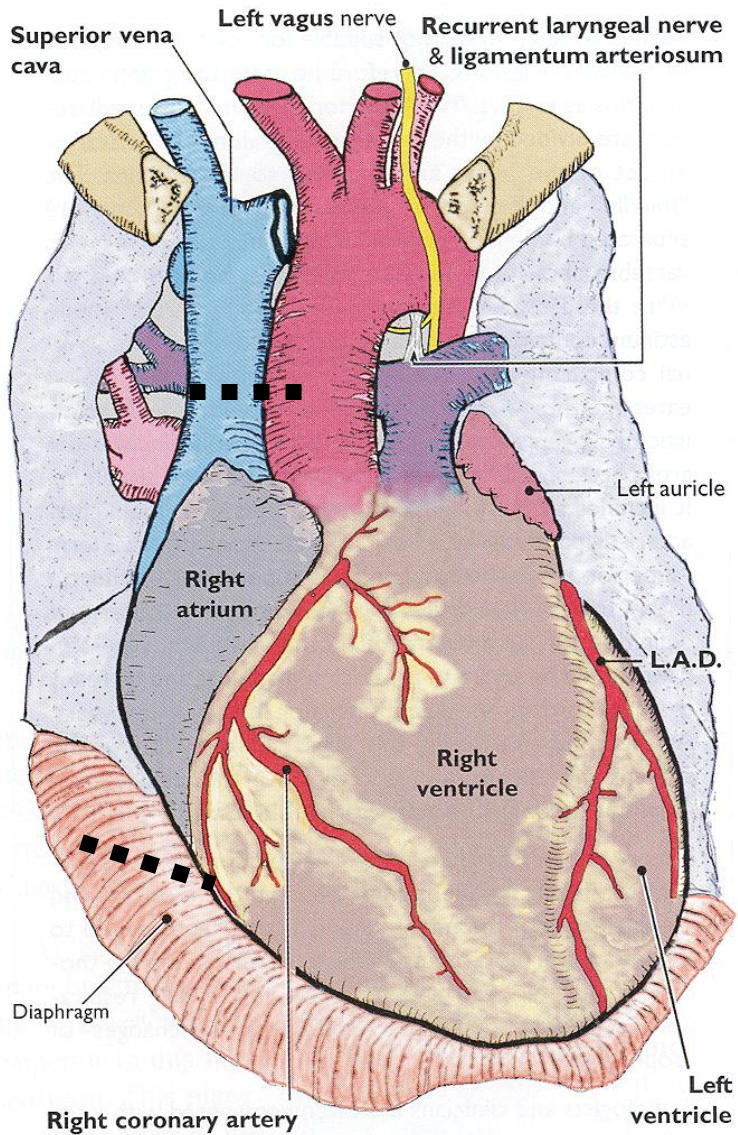
1. Cut across ascending aorta and pulmonary trunk



Removal of the heart



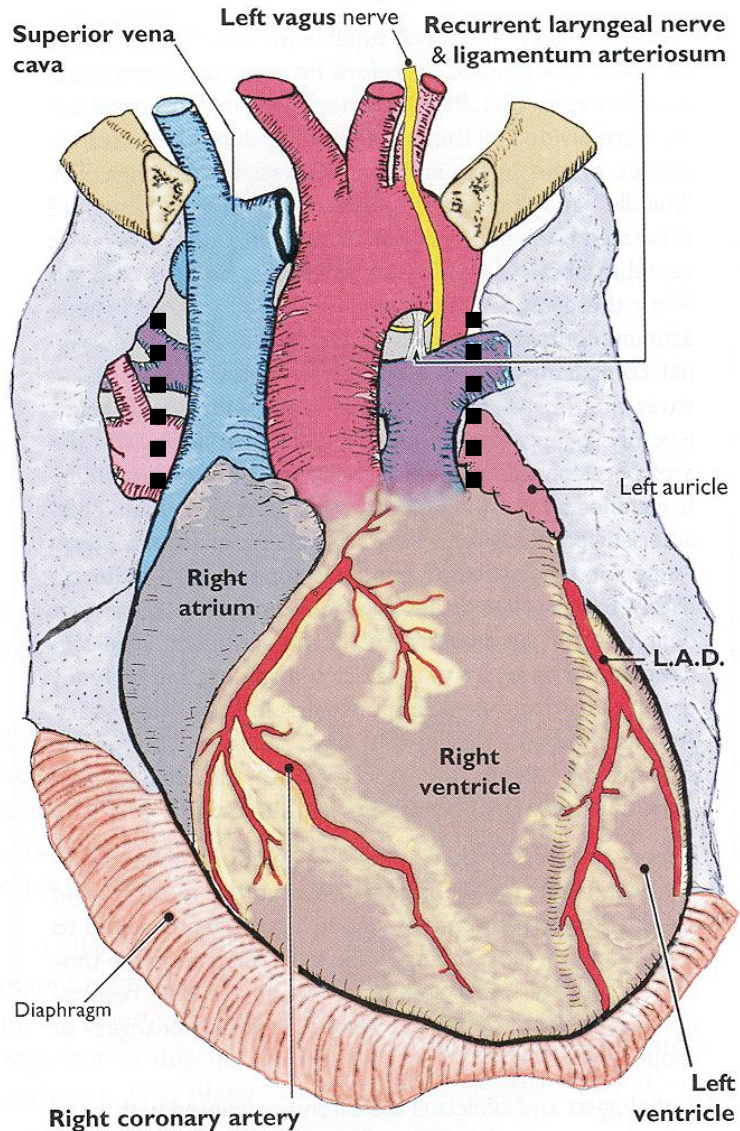
Removal of the heart



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Removal of the heart

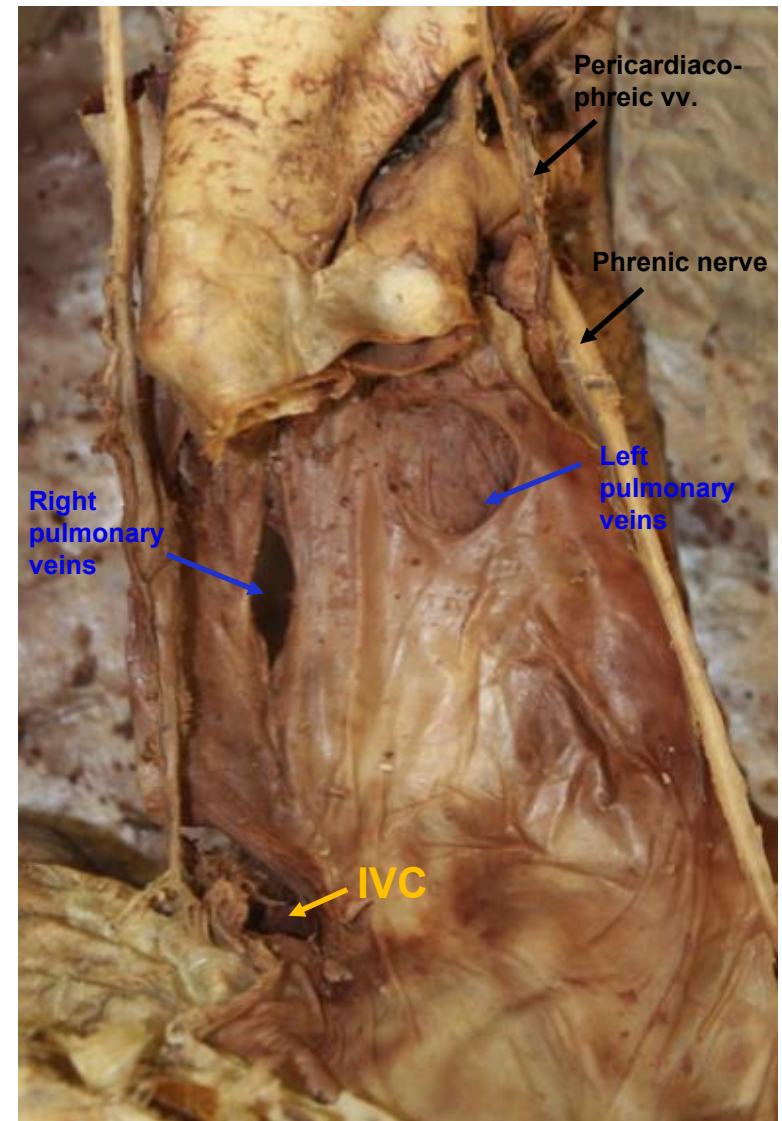
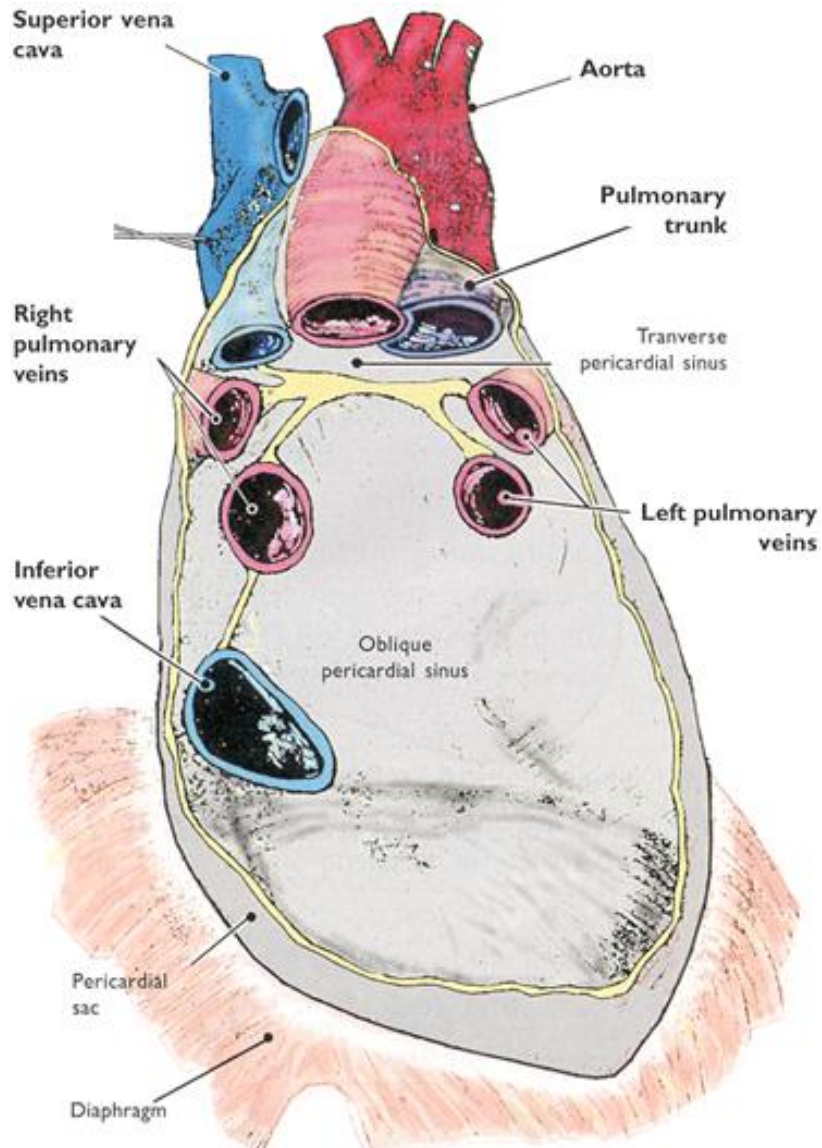


4. Lift the apex anteriorly and superiorly

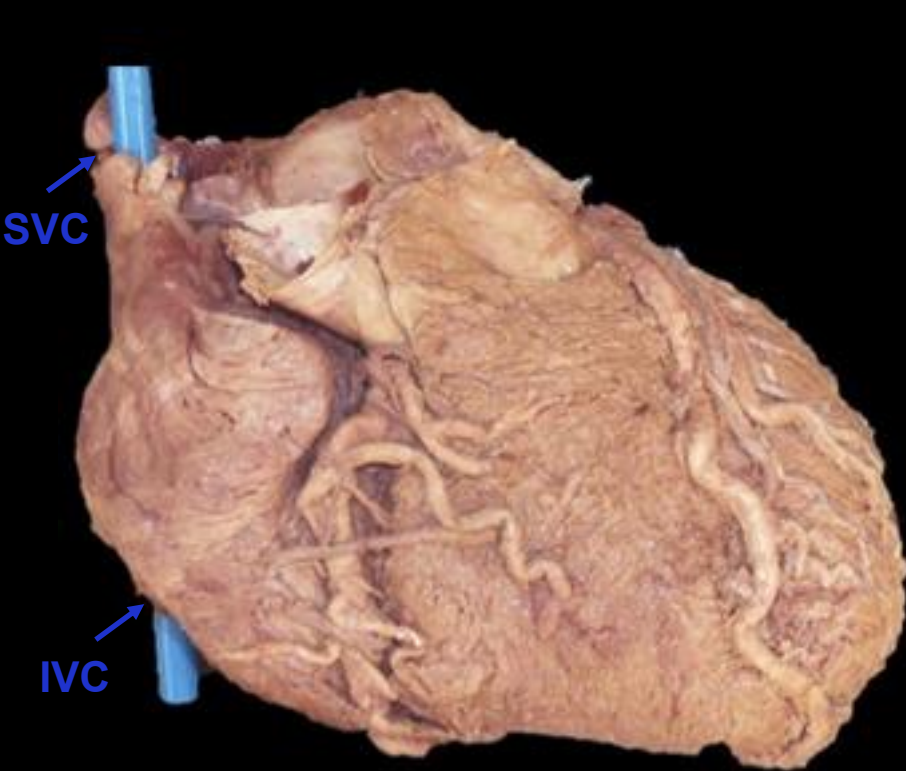
5. Cut across 4 pulmonary veins

6. Cut serous pericardium between transverse and oblique sinuses (posterior)

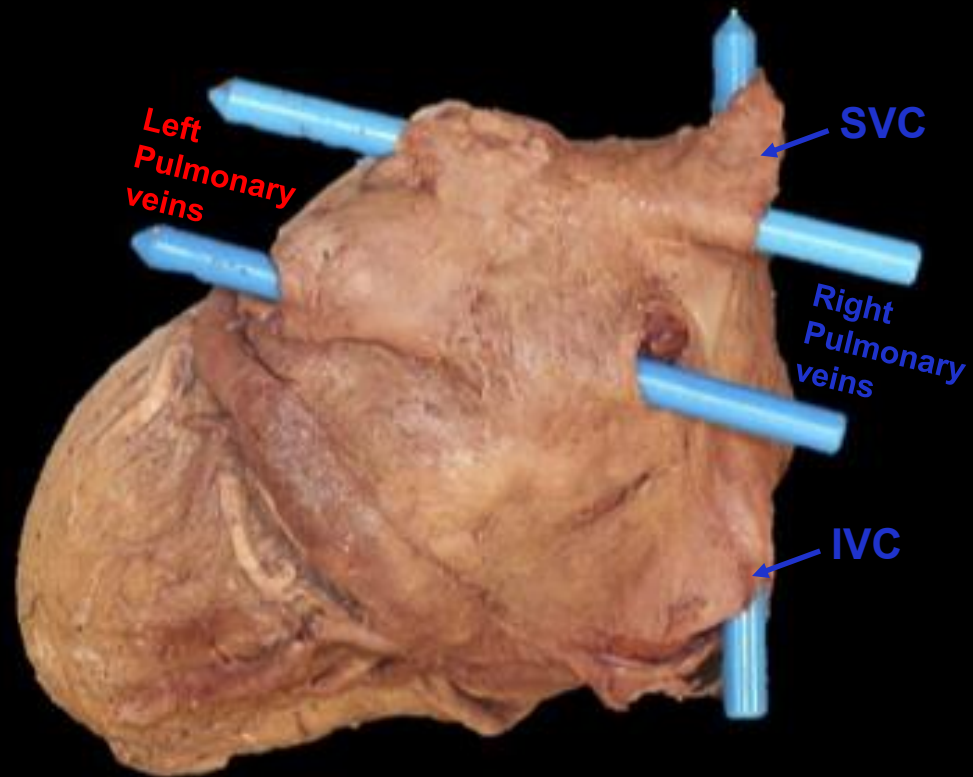
Removal of the heart



Anatomical position of the heart

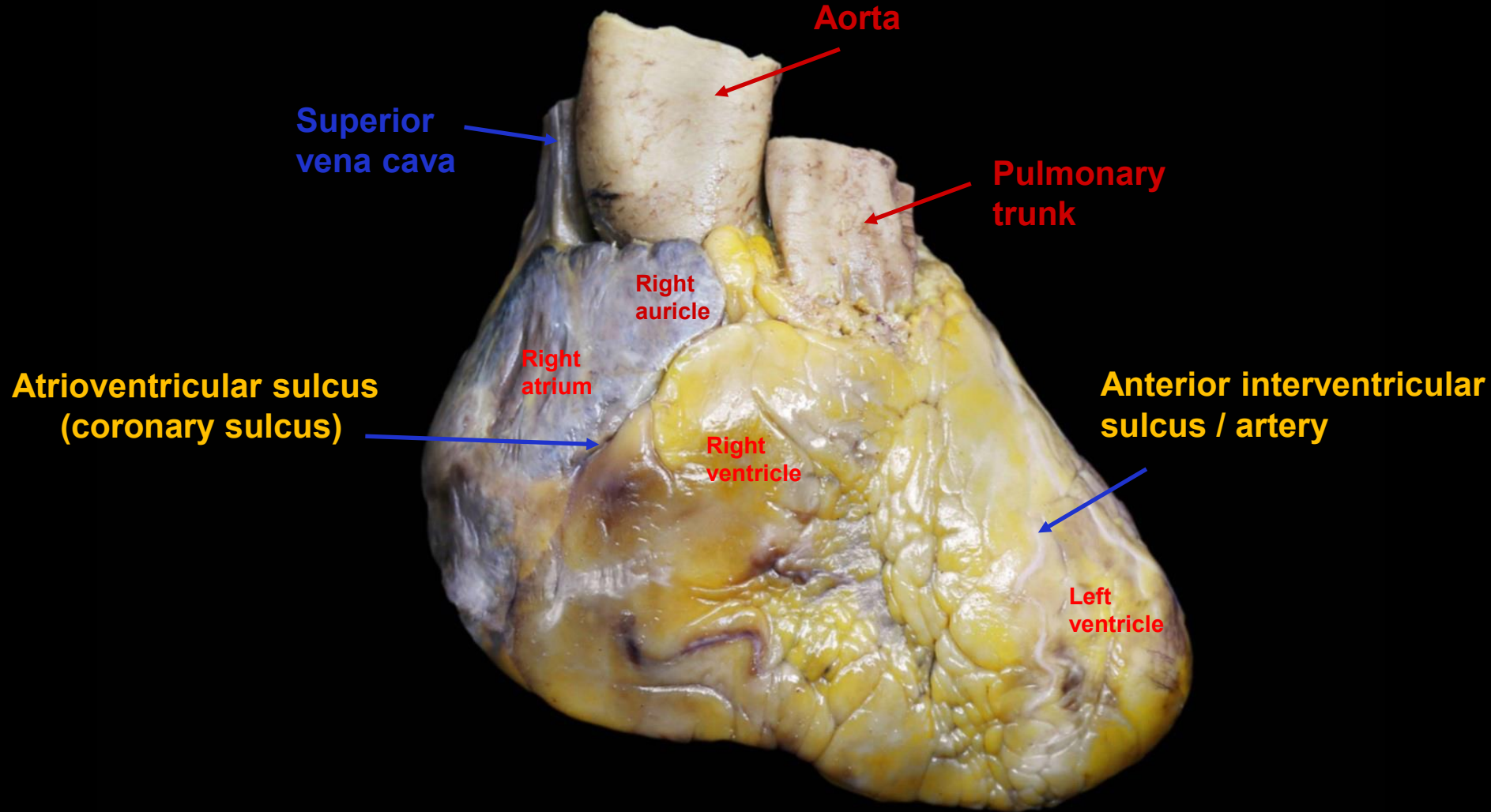


Anterior



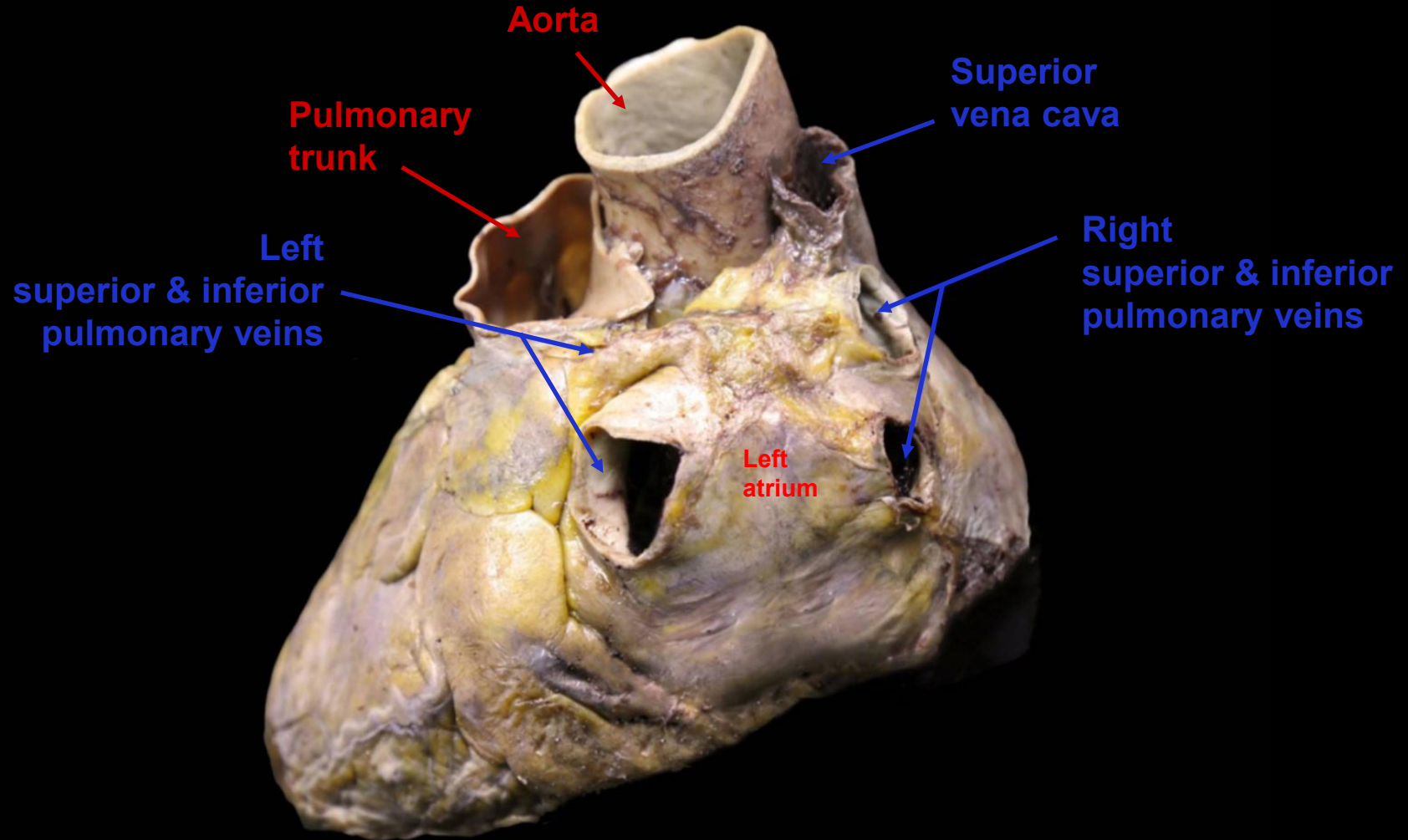
Posterior

Examine the gross features of the heart and great vessels



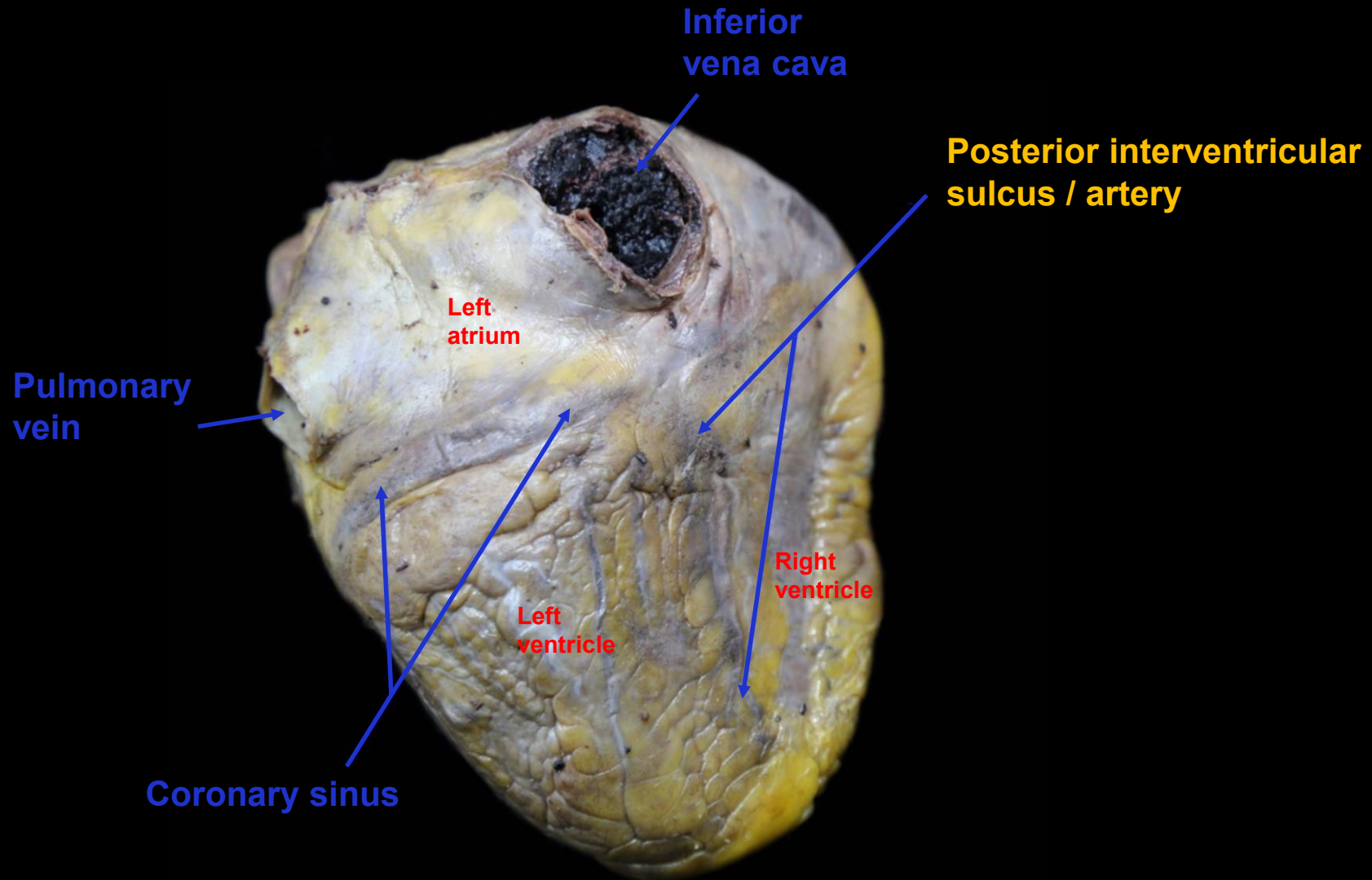
Heart, anterior

Examine the gross features of the heart and great vessels



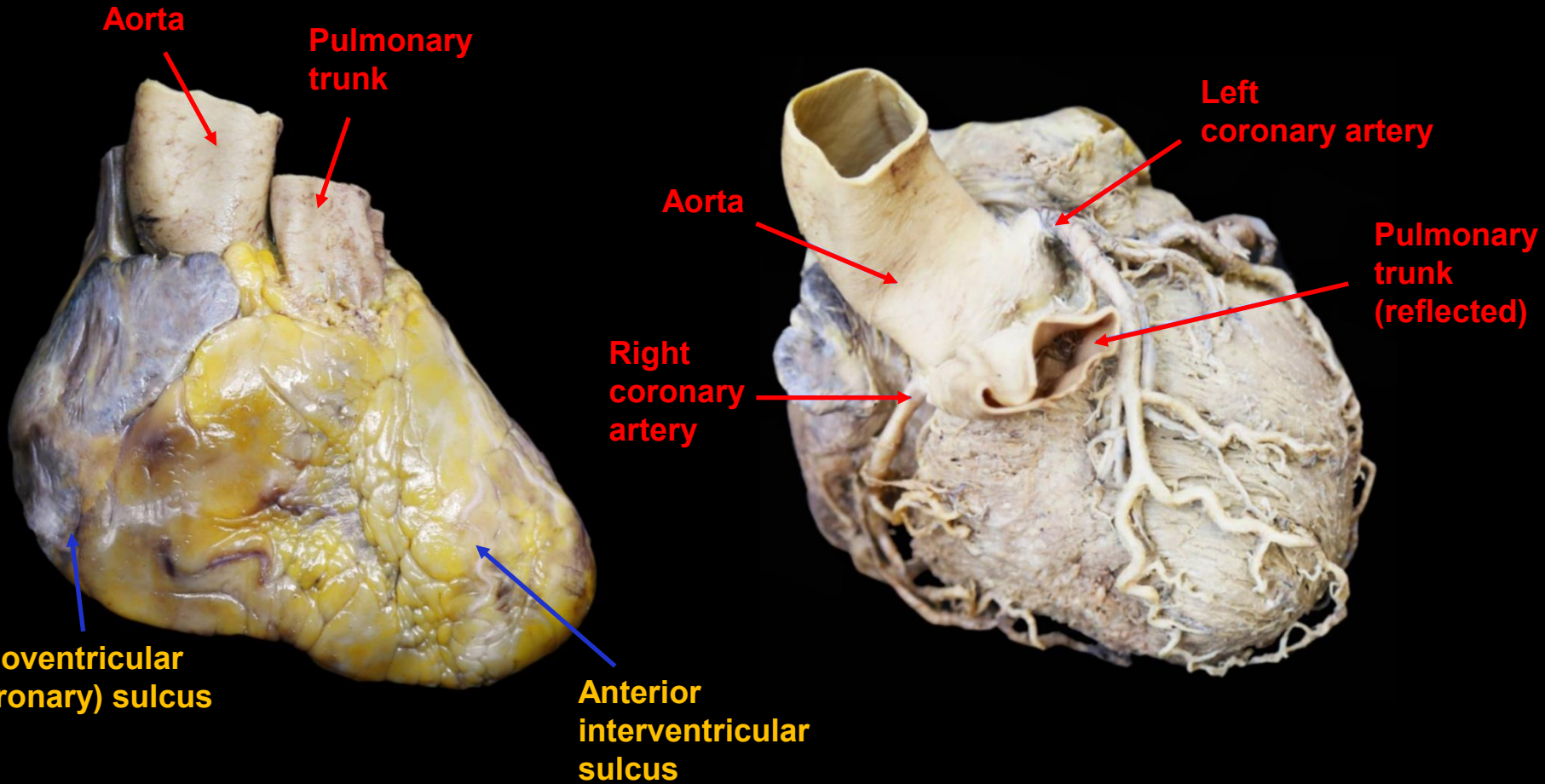
Heart, posterior

Examine the gross features of the heart and great vessels



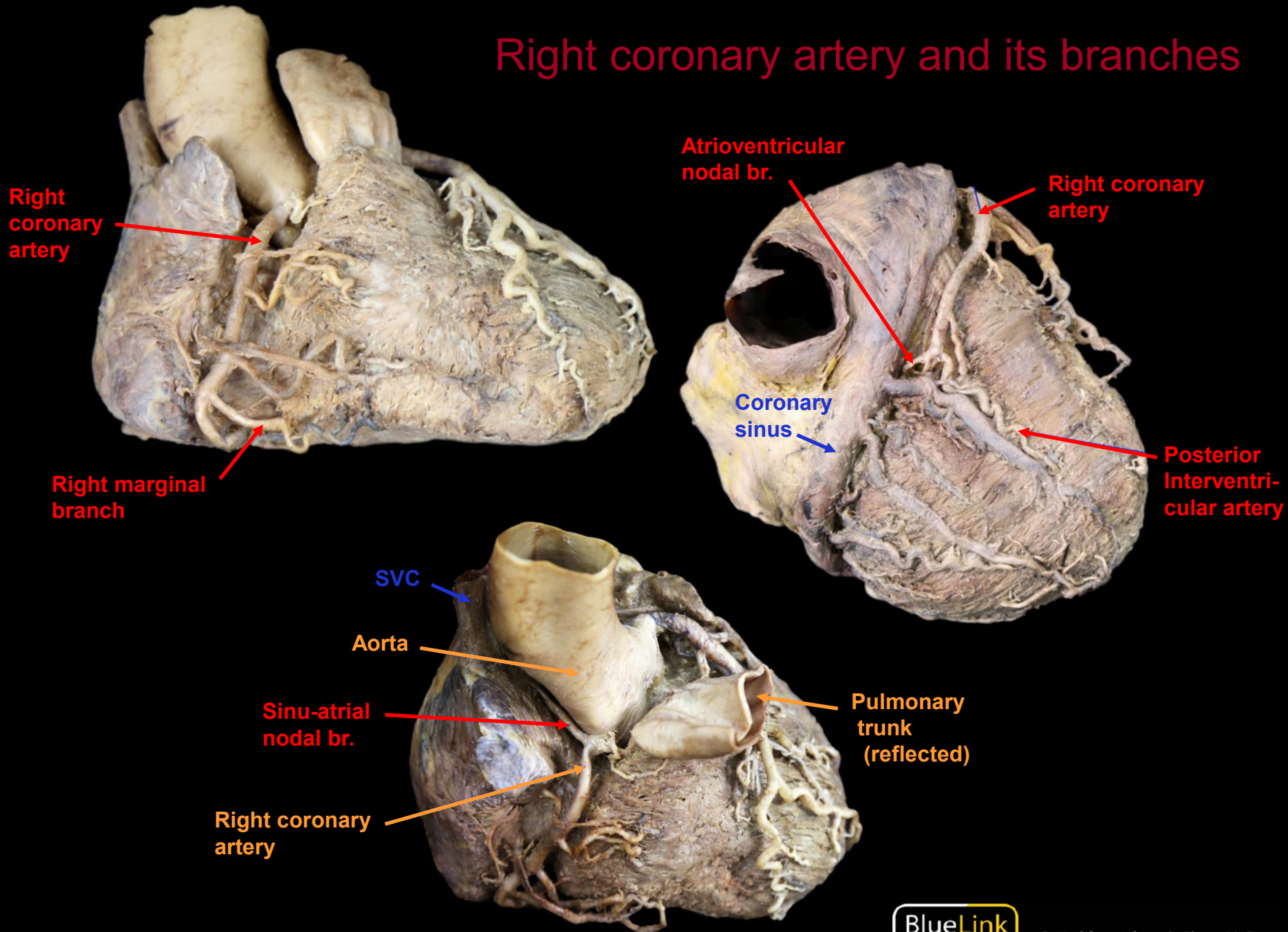
Heart, posteroinferior

Surface fat and coronary Arteries

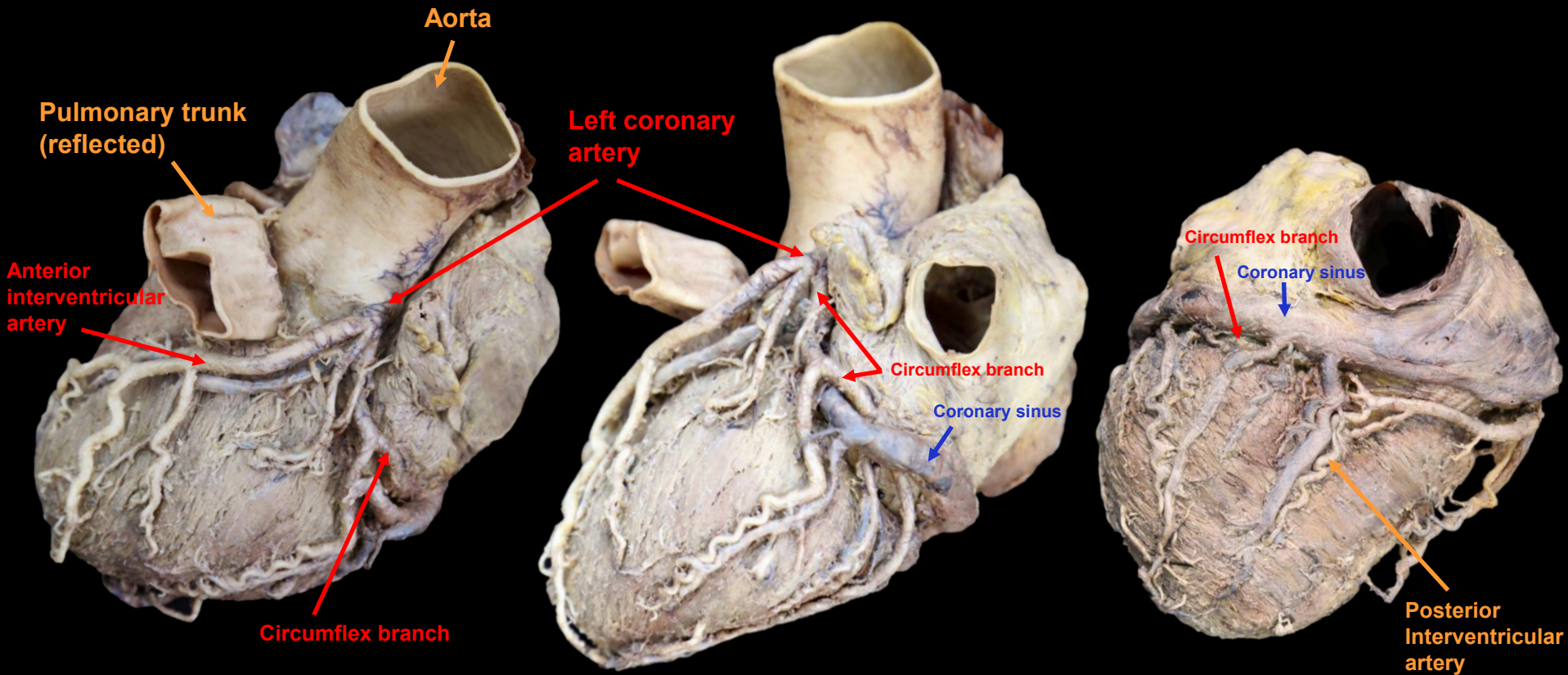


Heart, anterolateral

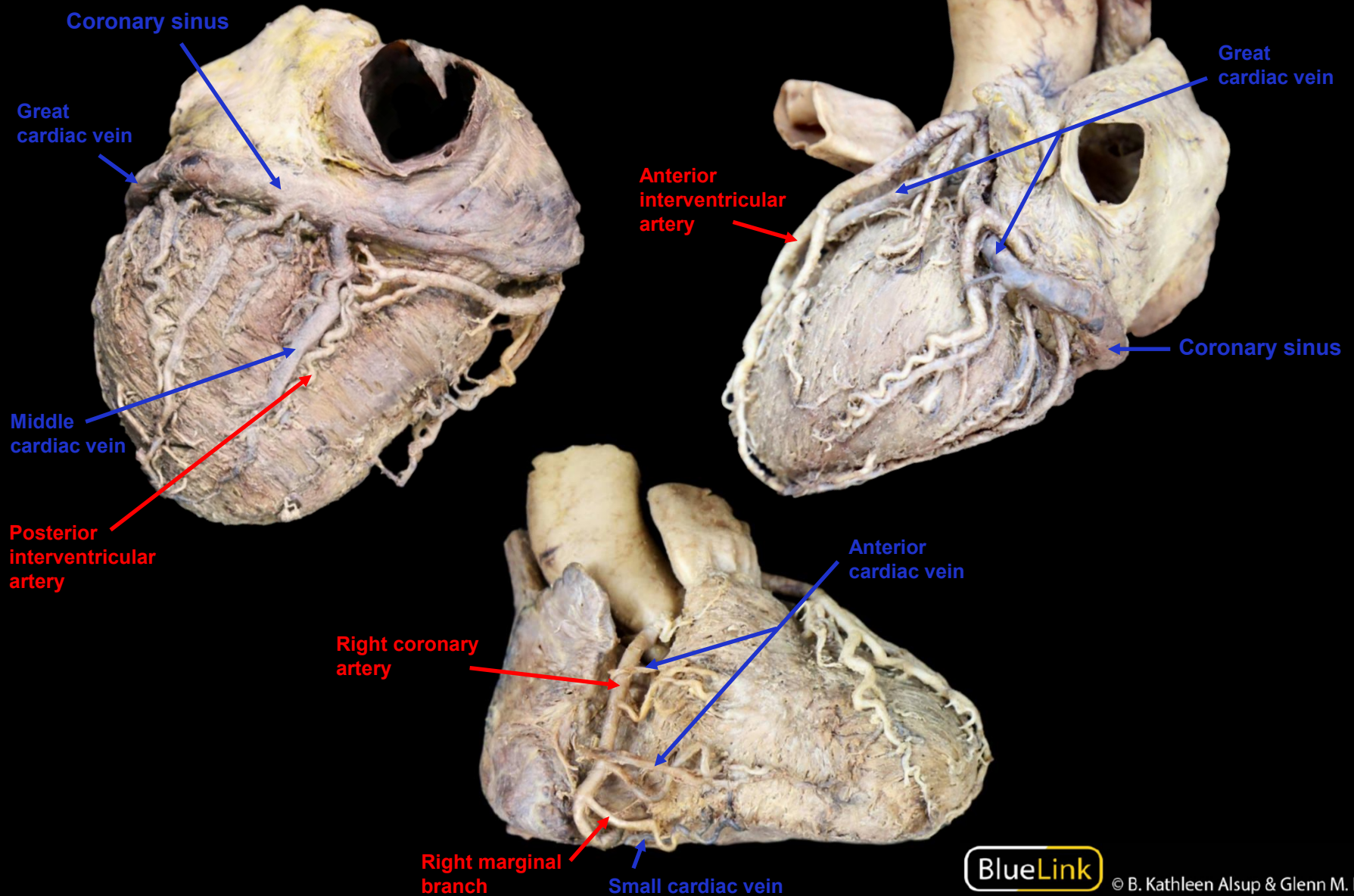
Right coronary artery and its branches



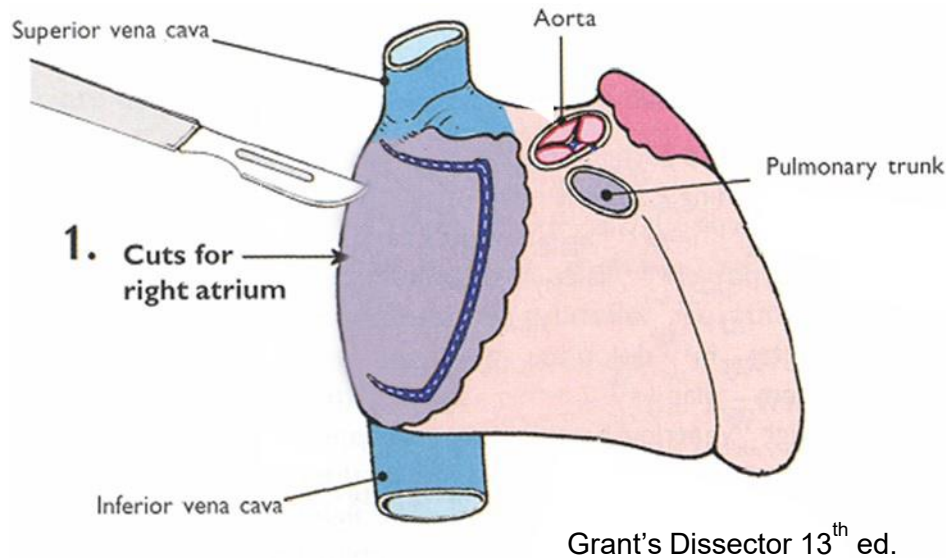
Left coronary artery and its branches



Examine venous drainage of the heart

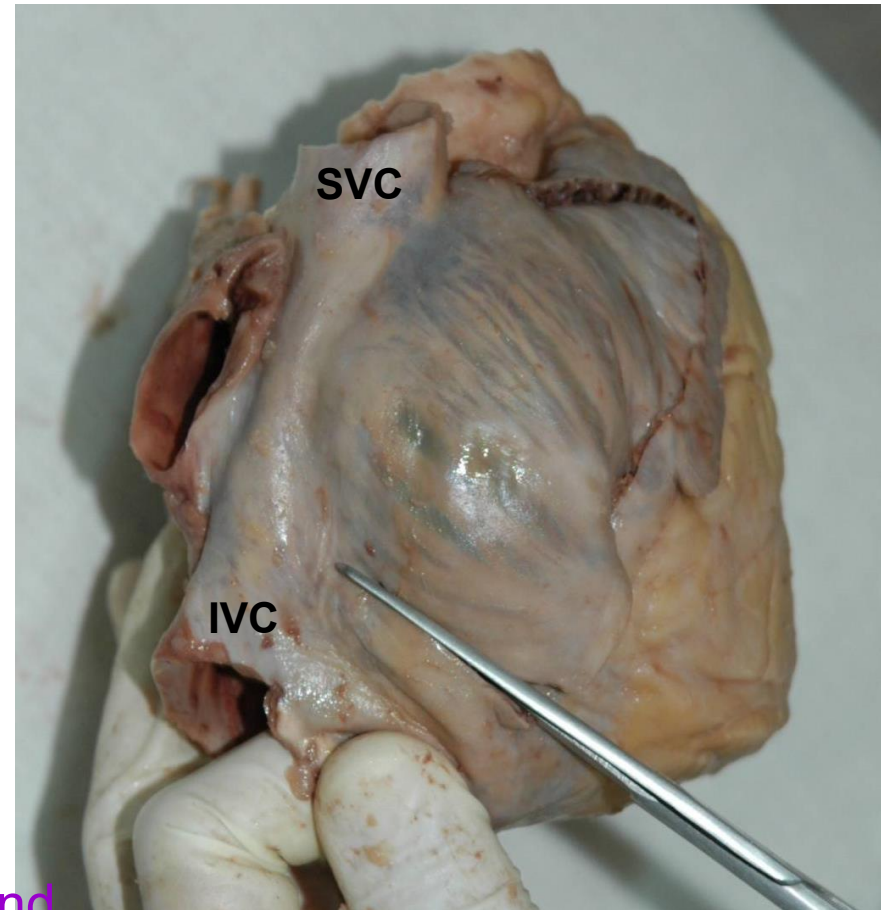


Opening anterior wall of the right atrium

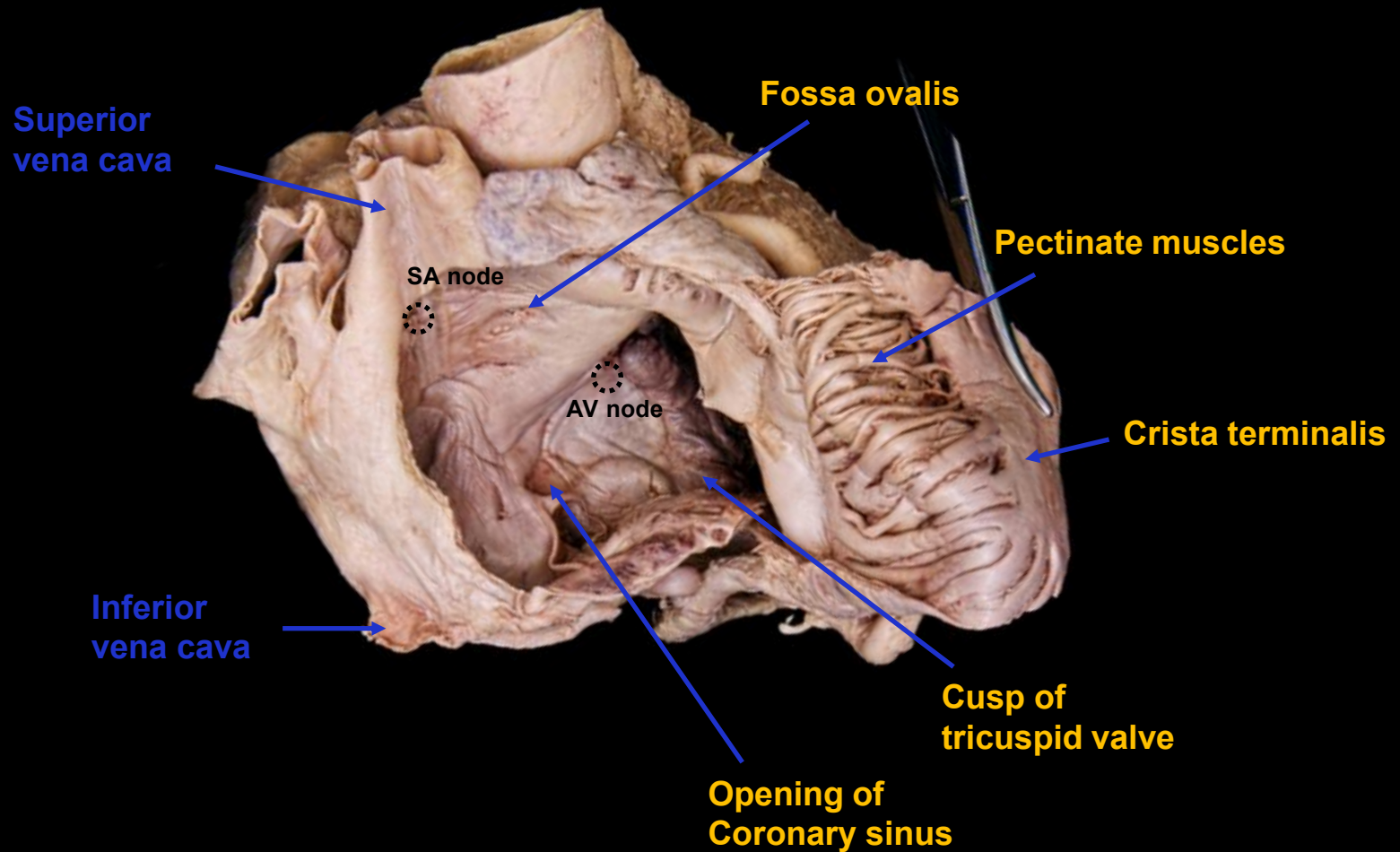


Make the following incisions:

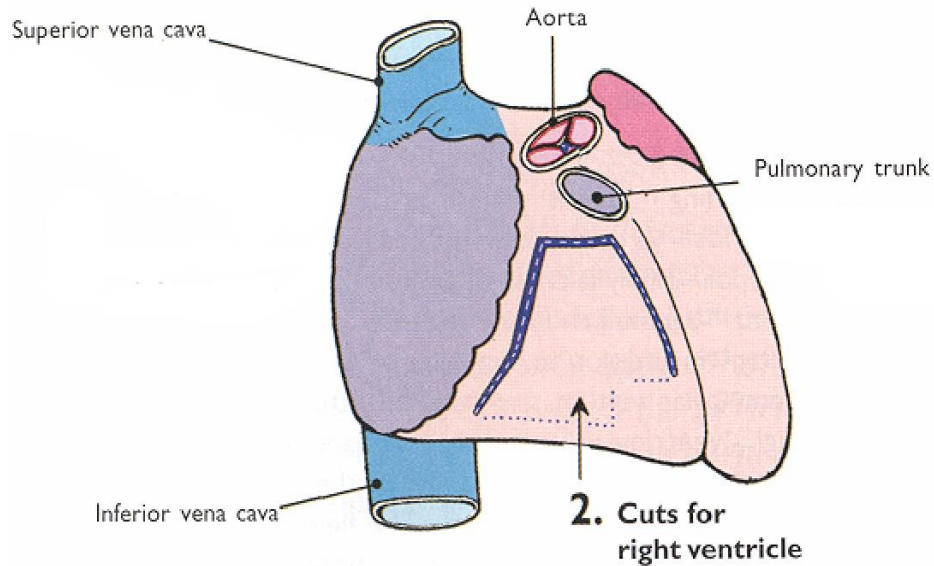
1. Horizontally, just below the entrance of the SVC into the right atrium
2. Vertically, cut about 1 cm away from and parallel to the atrioventricular sulcus
3. Horizontally, from second cut to the atrial-ventricular border
4. Reflect the flap of the anterior wall of the right atrium



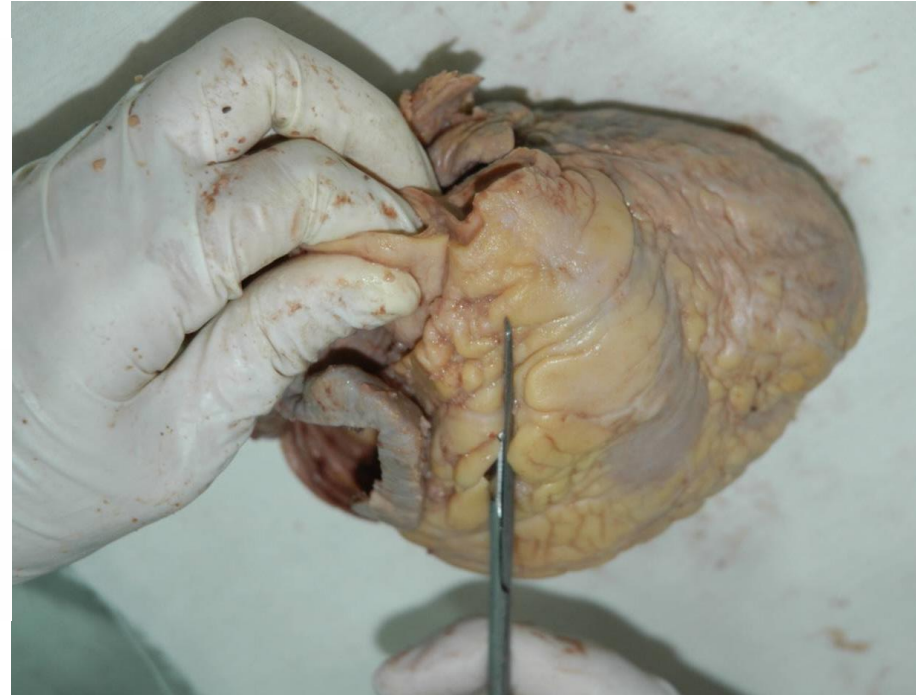
Examine the internal structures of the right atrium



Opening the right ventricle



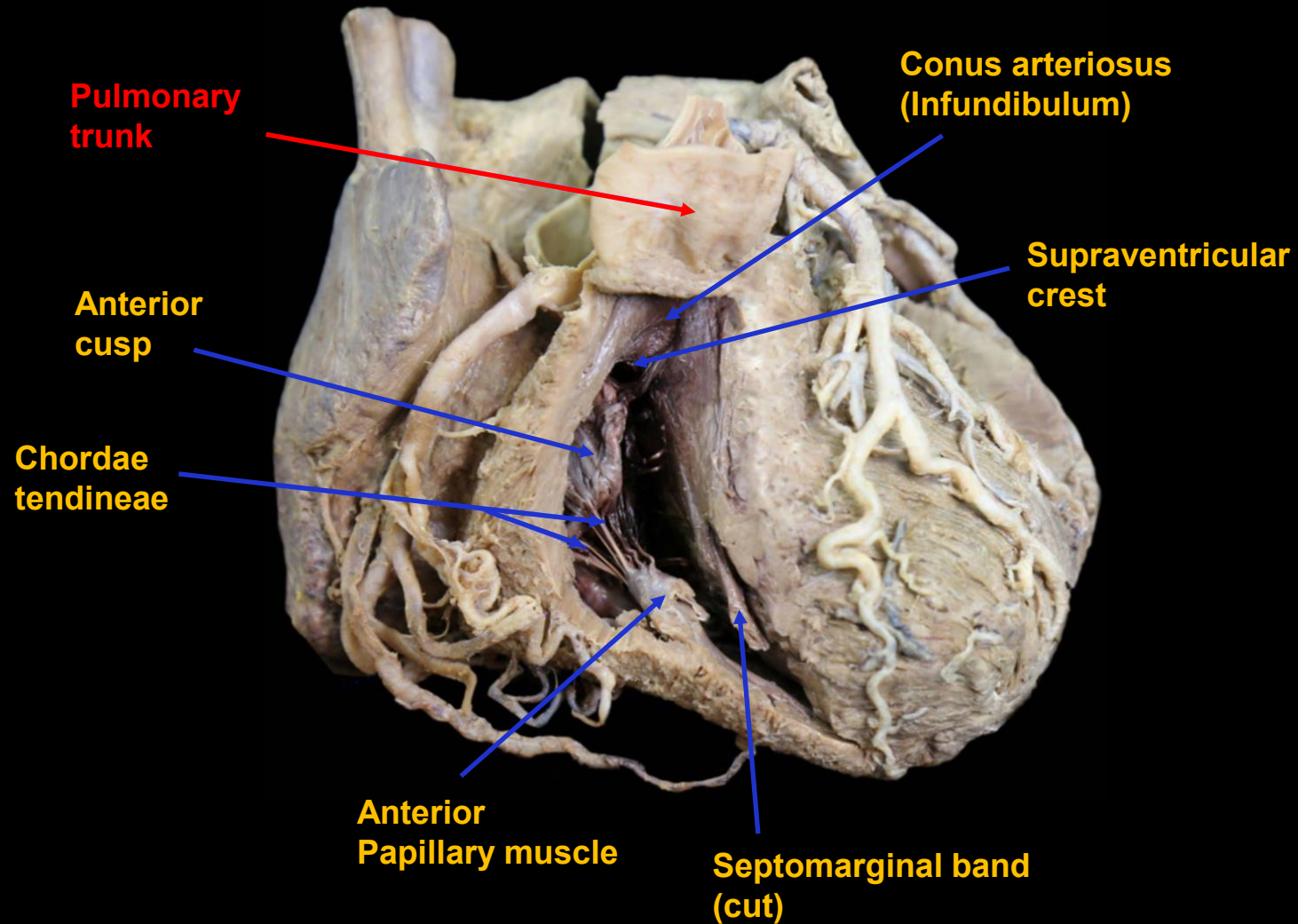
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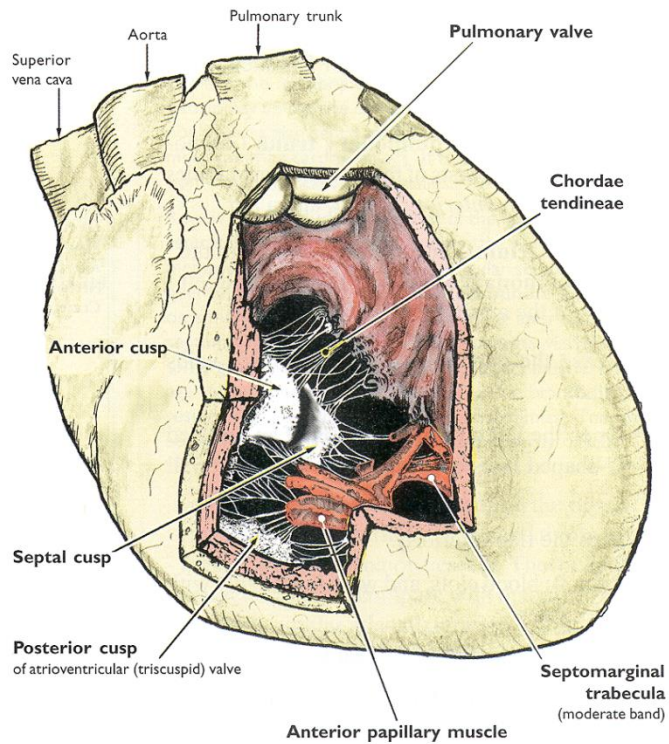
Make the following incisions:

1. Horizontally, below the area of the level of the pulmonary valve
2. Cut about 1 cm away from and parallel to the atrioventricular sulcus
3. Obliquely and medial to the anterior interventricular sulcus, where the LAD is found
4. Reflect the flap of the anterior wall of the right ventricle downward

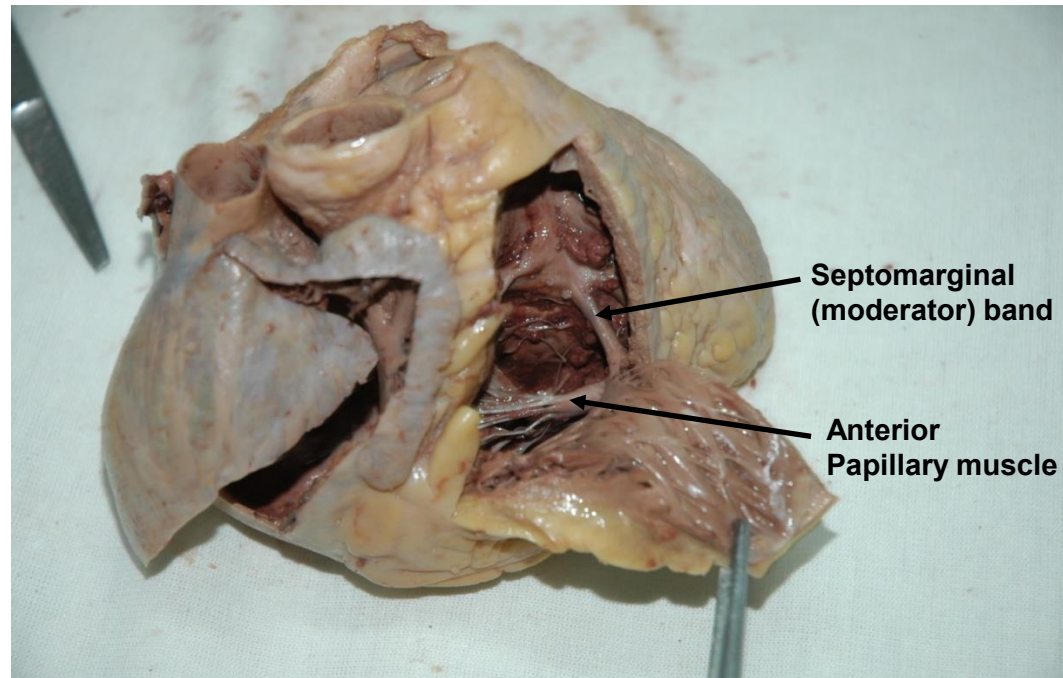
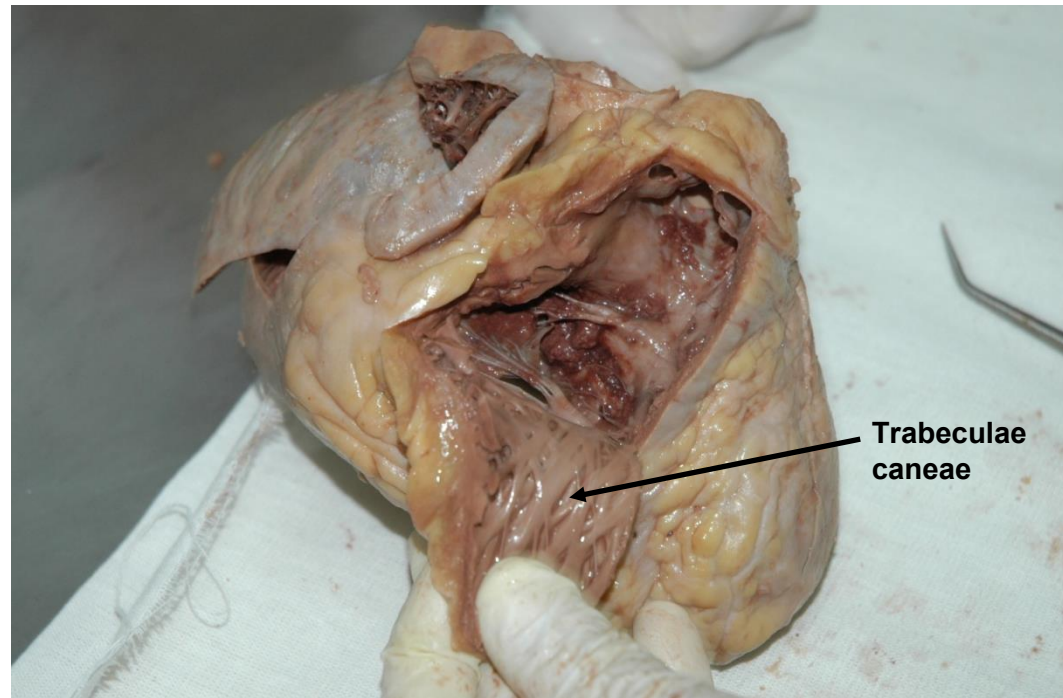
Examine the internal structures of the right ventricle



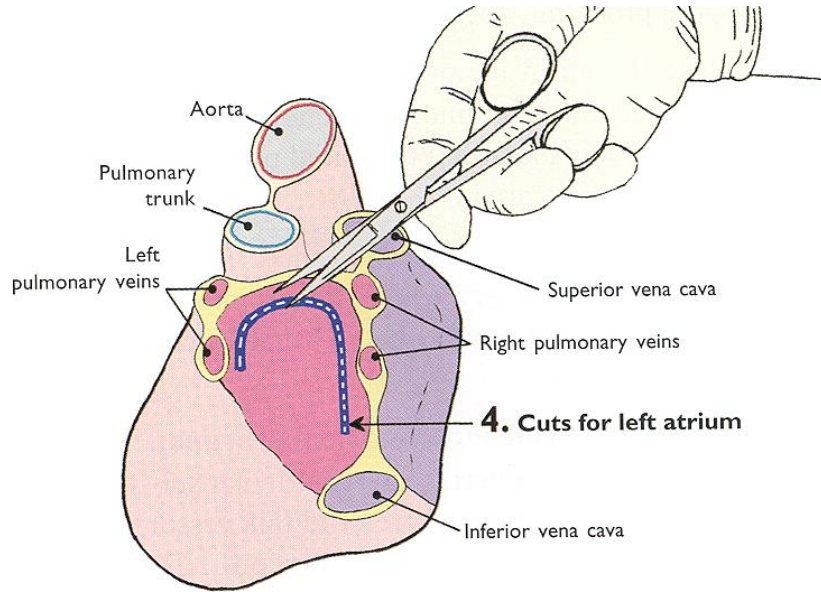
Right ventricle



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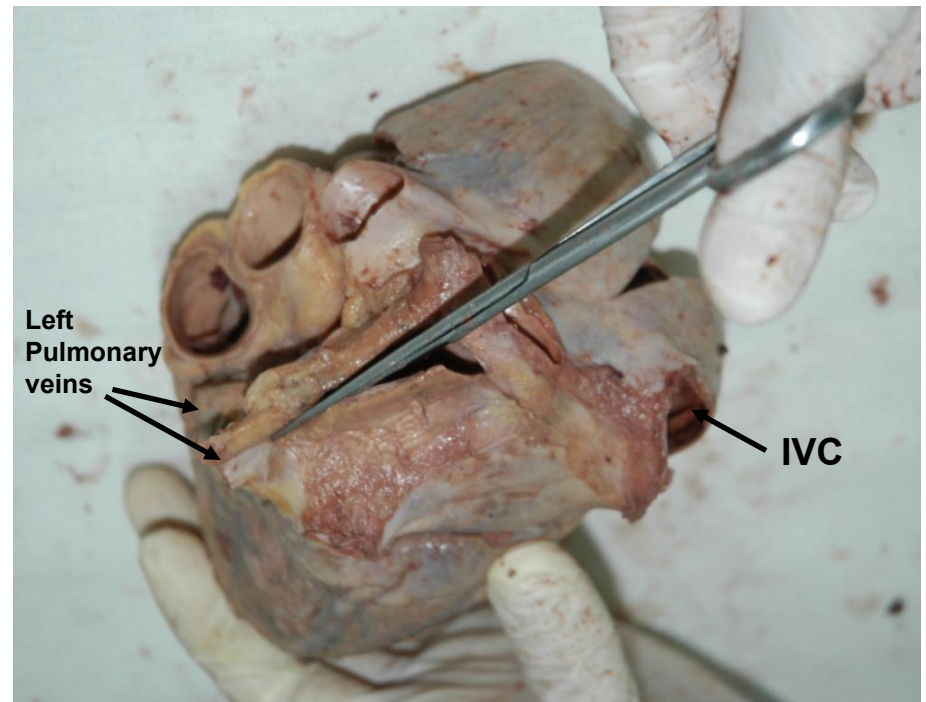
Opening the left atrium



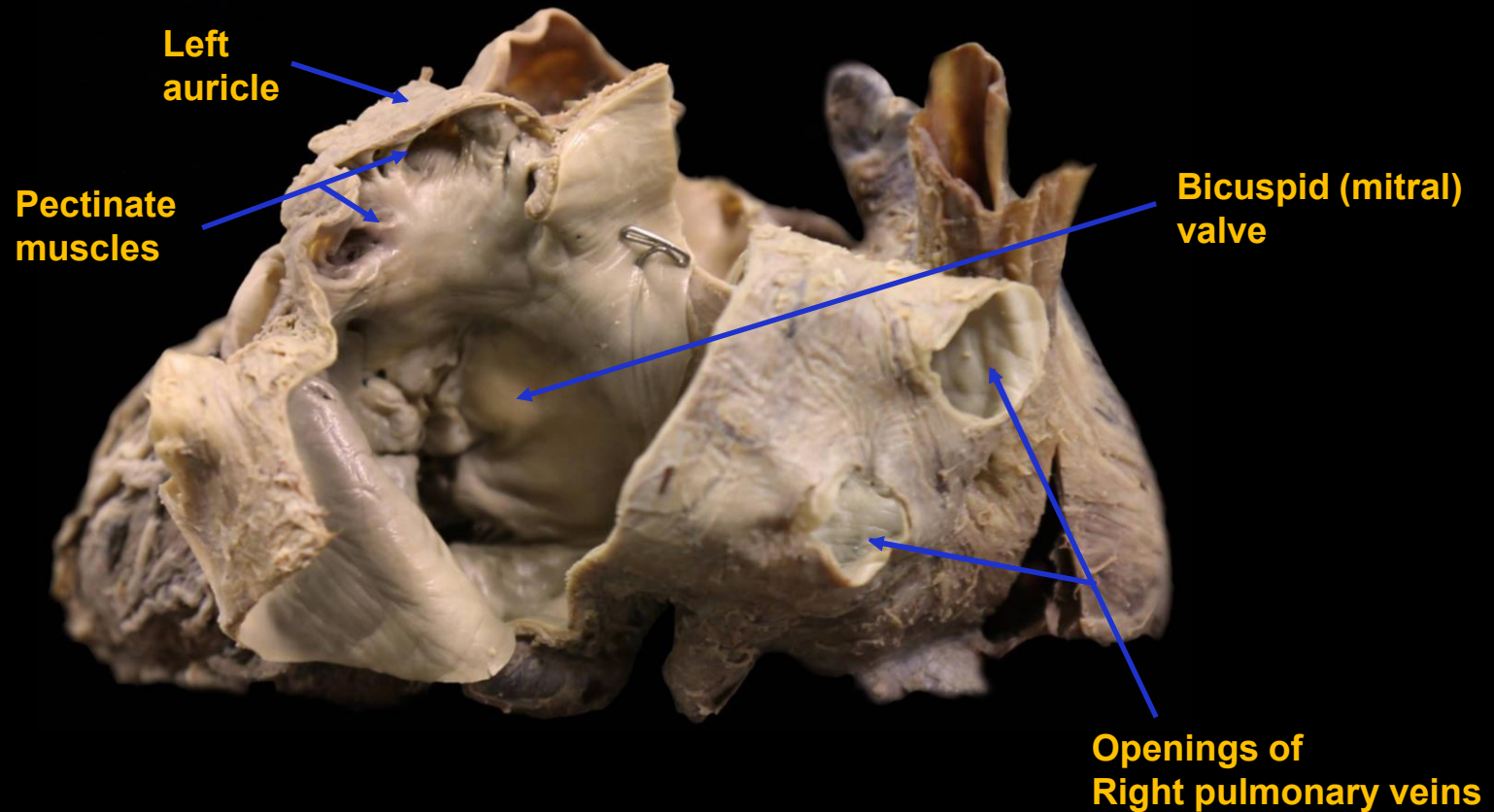
Grant's Dissector 13th ed.

Make the following incisions:

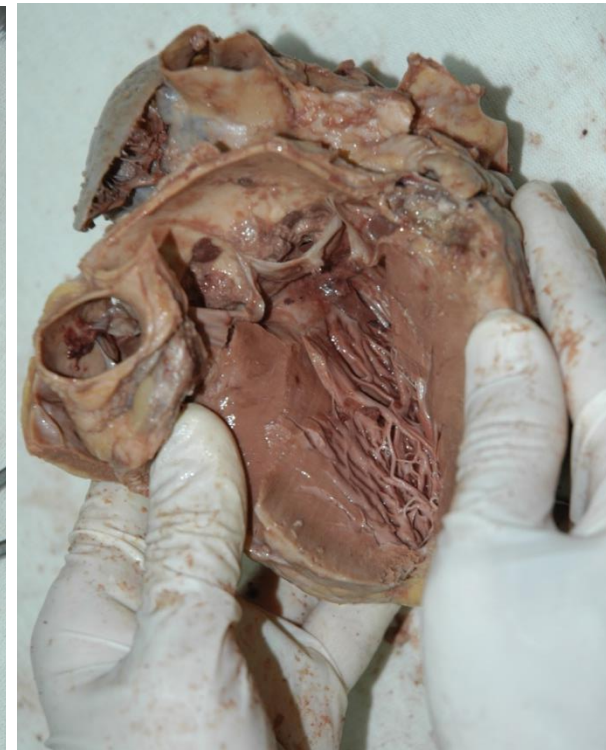
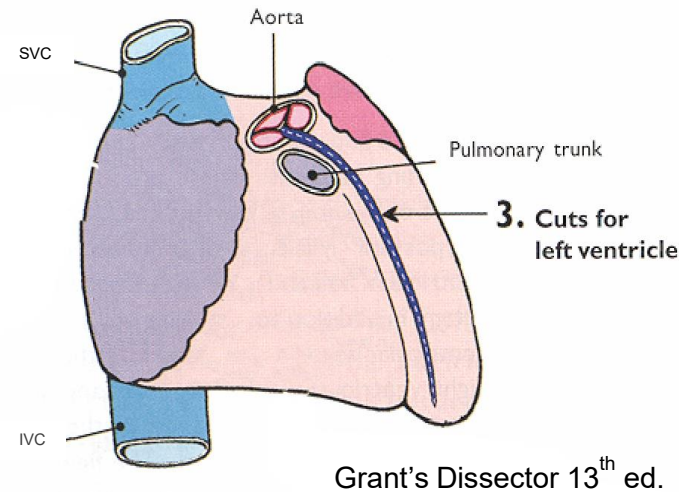
1. Cut an inverted U on the posterior wall of the left atrium.
2. Reflect the flap of tissue inferiorly
3. Cut along the left auricle



Examine the internal structures of the left atrium



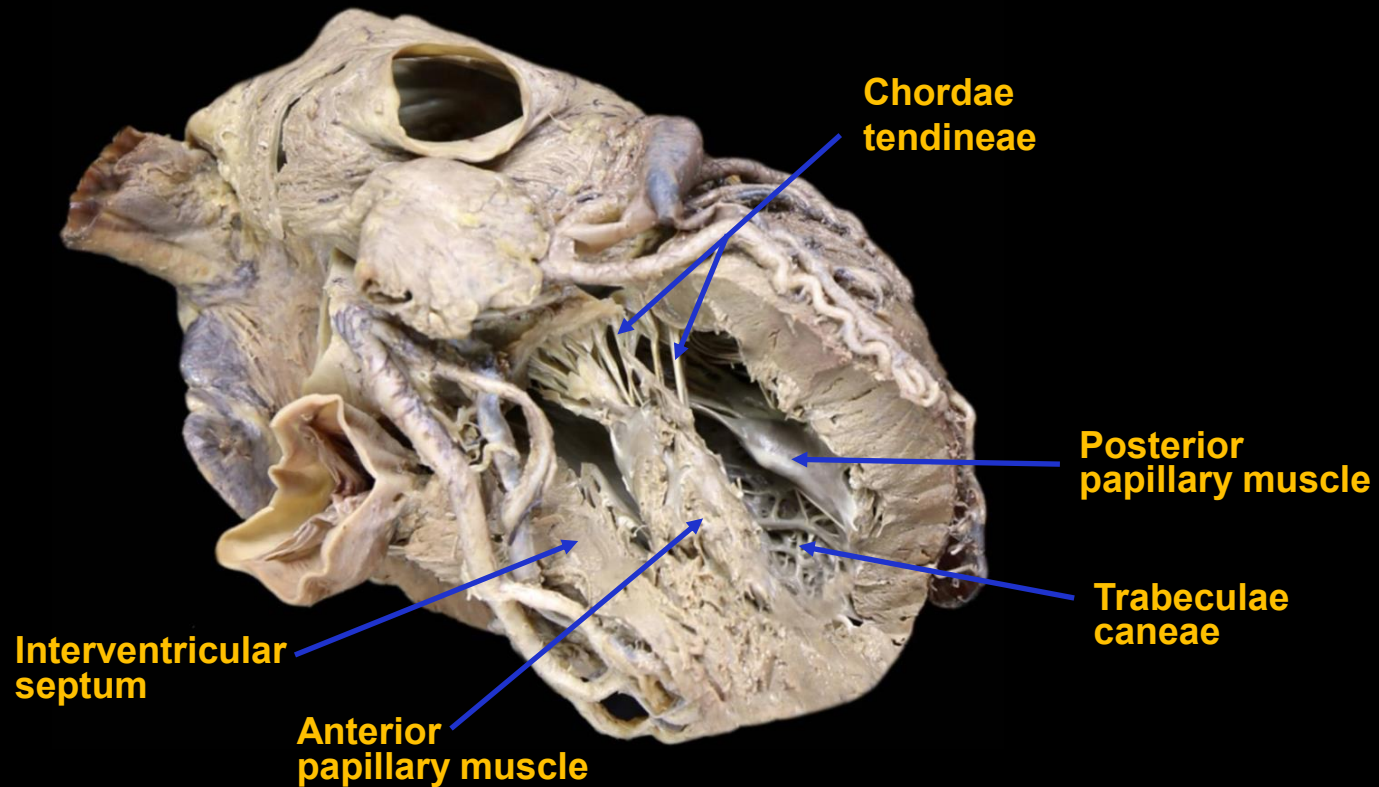
Opening the left ventricle



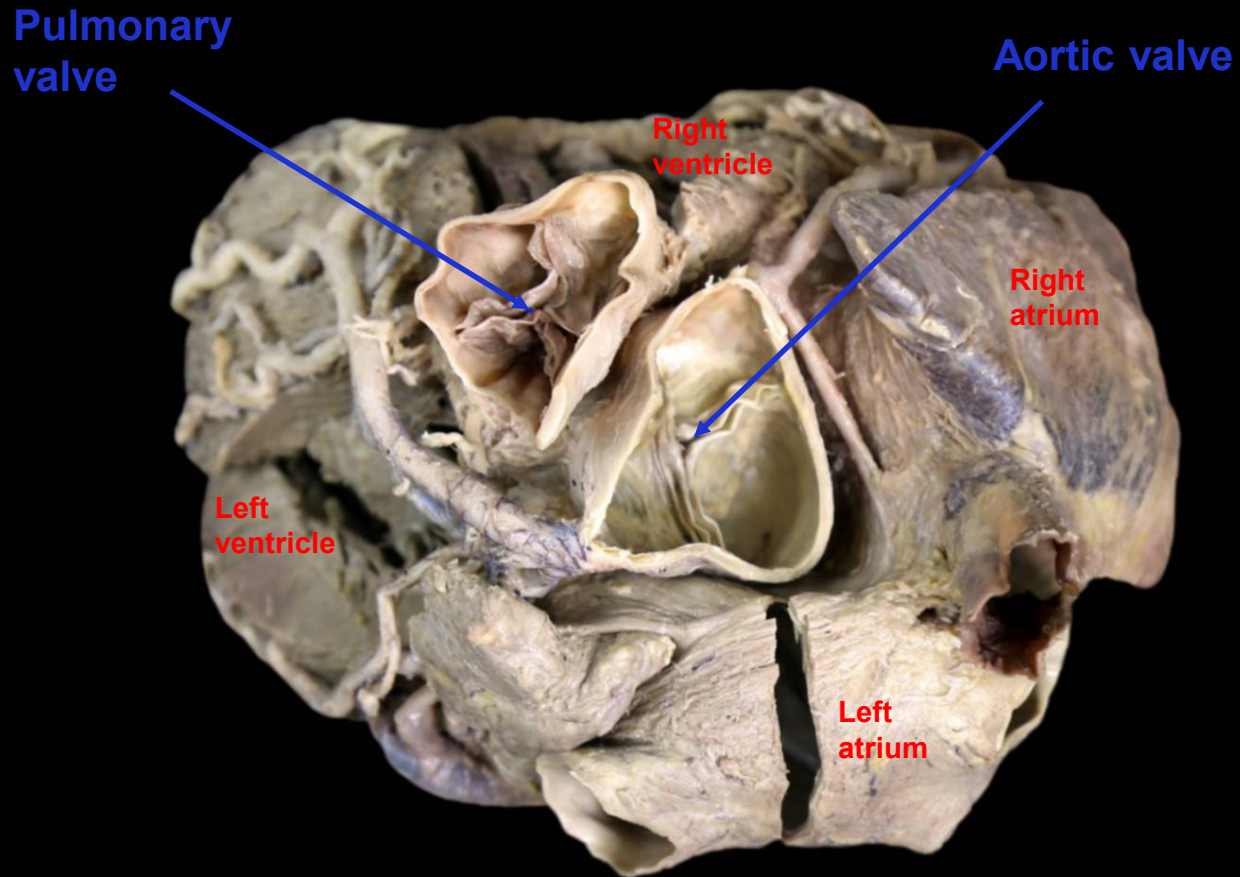
Make the following incisions:

1. Vertical cut 1 cm to the left and parallel to the anterior interventricular sulcus
2. Start the cut near the apex to just the root of the aorta
3. Extend the incision along the entire length of the aorta
4. Gently separate the two halves of the left ventricle

Examine the internal structures of the left ventricle

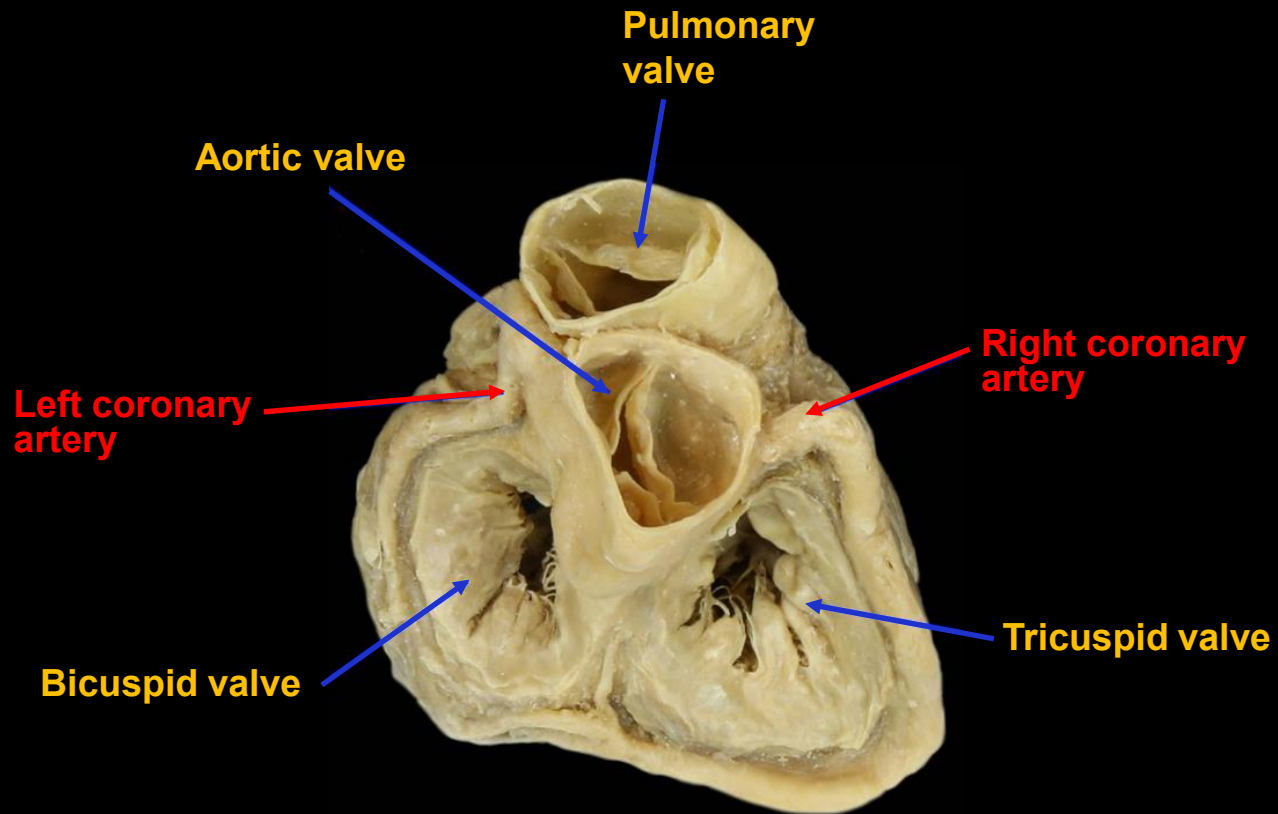


Examine the heart valves



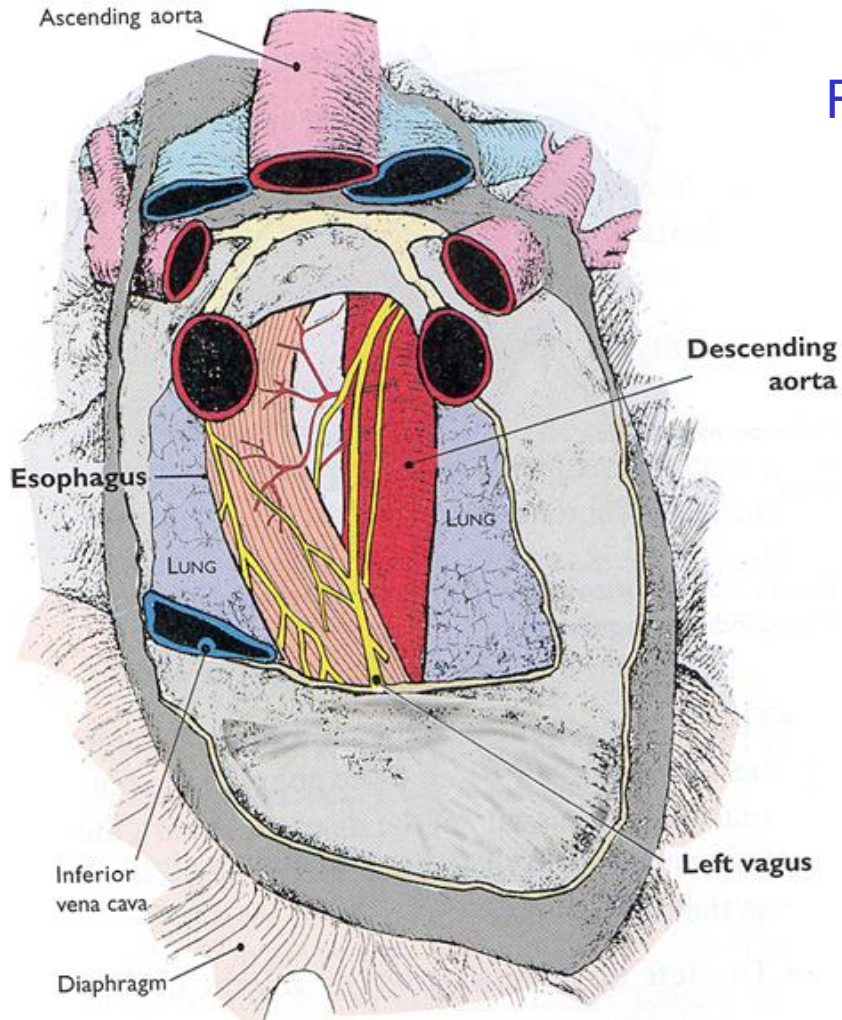
Heart, superior

Examine the heart valves

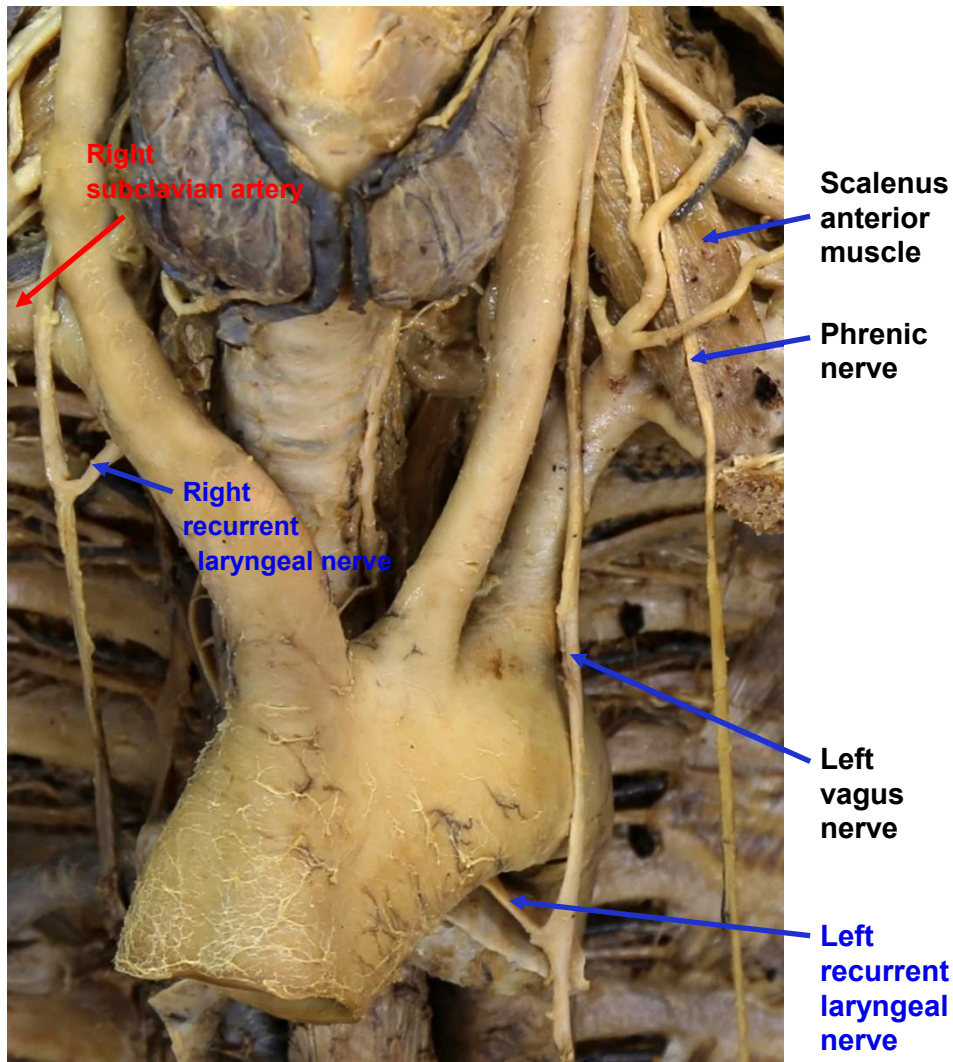


Posterior mediastinum

Remove the remainder of the pericardial sac, identify and clean

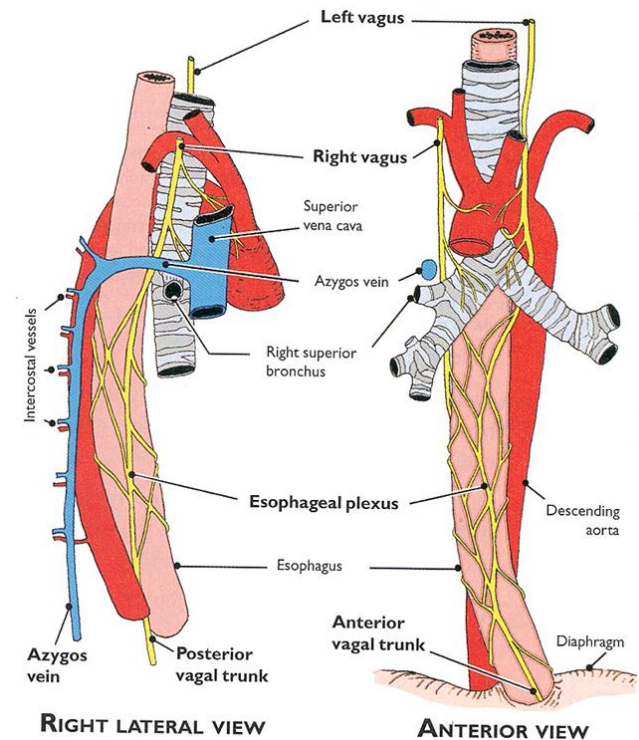


Posterior mediastinum



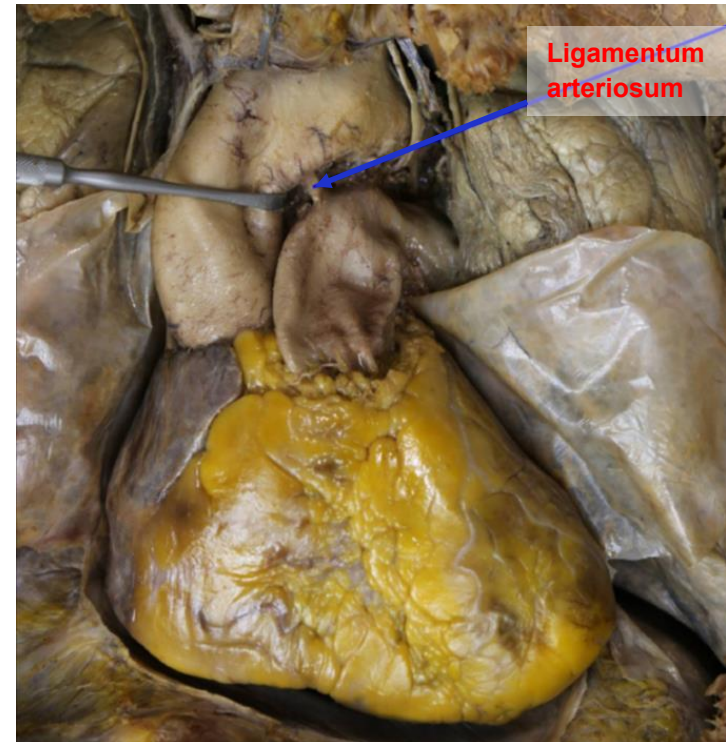
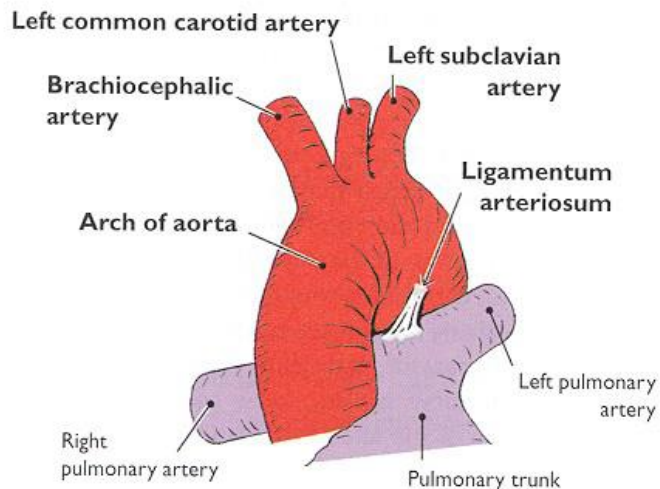
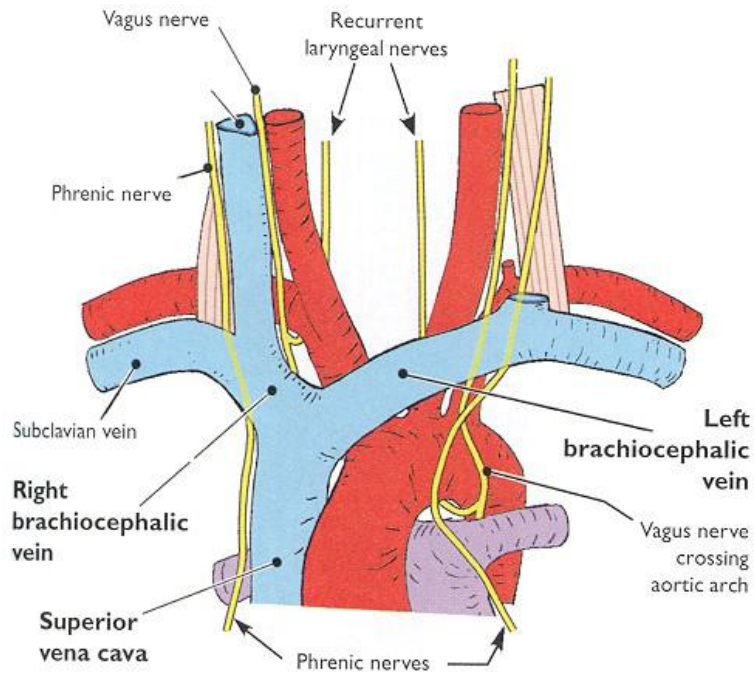
Remove the remainder of the pericardial sac, identify and clean

1. **Right vagus nerve** posterior to the root of right lung and spread out as esophageal plexus
2. **Left vagus nerve** as it crosses the left side of the aortic arch



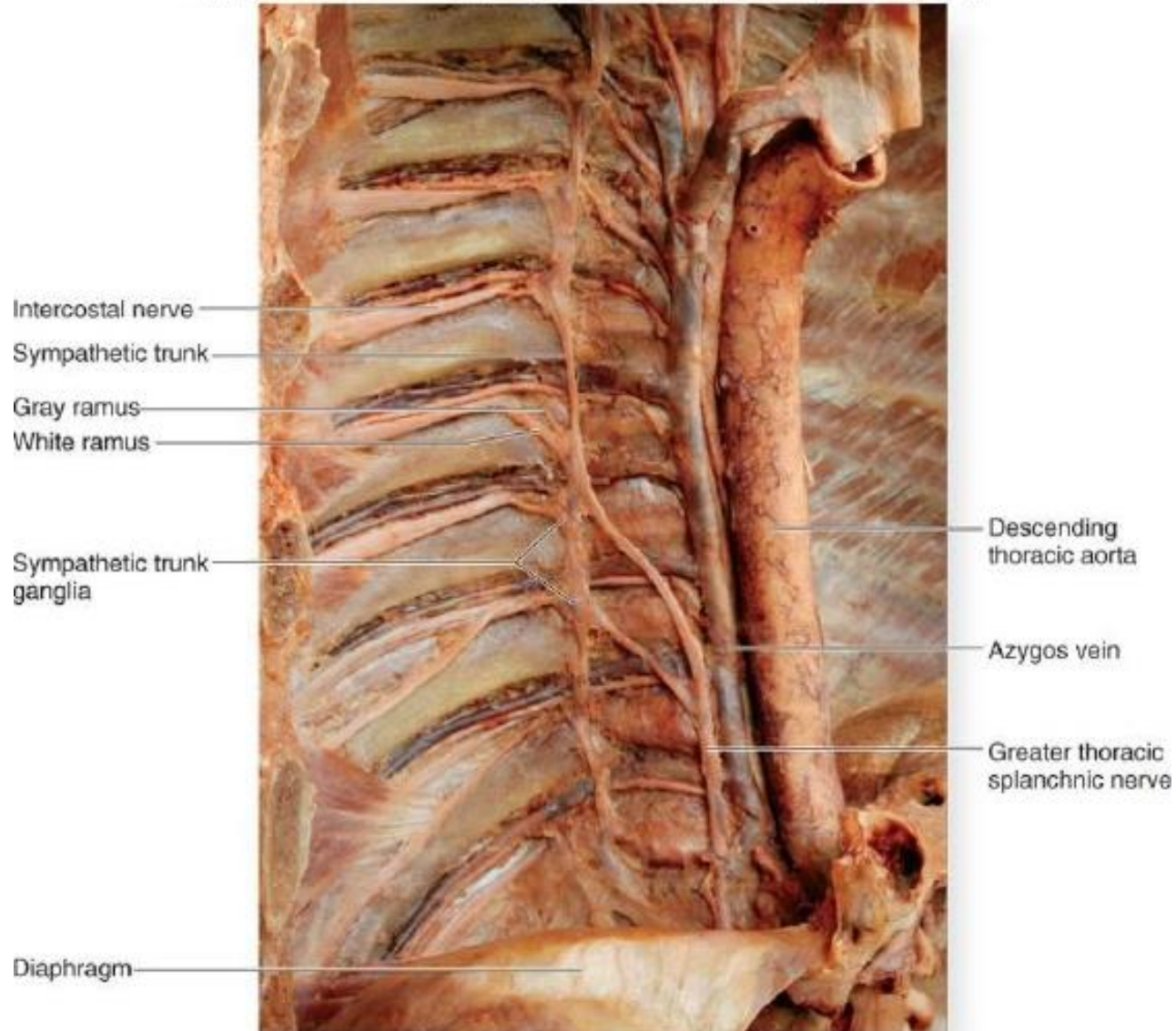
Posterior mediastinum

3. **Ligamentum arteriosum**, aortic arch is connected to the left pulmonary artery
4. **Left recurrent laryngeal nerve**, as it courses immediately posterior to ligamentum arteriosum



Posterior mediastinum

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THE END